

JEE Advanced Exam 2019
(Paper & Solution)
PAPER-2

PART-I (PHYSICS)

SECTION – 1(Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks: **+4** If only (all) the correct option(s) is (are) chosen.

Partial Marks:**+3** If all the four options are correct but **ONLY** three options are chosen.

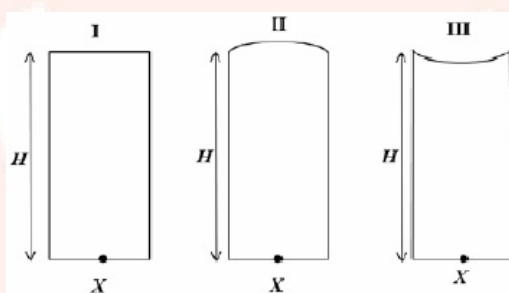
Partial Marks: **+2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.

Partial Marks: **+1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks: **0** If none of the option is chosen (i.e. the question is unanswered).

Negative Marks: **–1** In all other cases.

1. Three glass cylinders of equal height $H = 30$ cm and same refractive index $n = 1.5$ are placed on a horizontal surface as shown in figure. Cylinder I has a flat top, cylinder II has a convex top and cylinder III has a concave top. The radii of curvature of the two curved tops are same ($R = 3$ m). If H_1, H_2 and H_3 are the apparent depth of a point X on the bottom of the three cylinders, respectively, the correct statement (s) is / are –



- (1) $0.8 \text{ cm} < (H_2 - H_1) < 0.9 \text{ cm}$ (2) $H_3 > H_1$
 (3) $H_2 > H_3$ (4) $H_2 > H_1$

Soln.: [3, 4]

For fig. I

It is given that its top is flat or plane. So, $R = \infty$ for plane surface,

As we know that

$$\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_2 - \mu_1}{R} \quad \text{Here } \mu_2 = 1, \mu_1 = 1.5, u = -30 \text{ cm}, v = -H_1$$

$$\frac{1}{-H_1} - \frac{1.5}{-30} = \frac{1 - 1.5}{\infty}$$

$$H_1 = \frac{30}{1.5} = 20 \text{ cm}$$

For fig. II,

Applying following formula

$$\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here $\mu_2=1$, $\mu_1=1.5$, $u = -30$ cm, $v = -H_2$, $R = -3m = -300$ cm

$$\frac{1}{-H_2} - \frac{1.5}{-30} = \frac{1-1.5}{-300}$$

$$\frac{1}{-H_2} + \frac{15}{300} = \frac{0.5}{300} = \frac{1}{600}$$

$$\frac{1}{H_2} = \frac{15}{300} - \frac{1}{600} = \frac{30-1}{600} = \frac{29}{600}$$

$$H_2 = 20.6 \text{ cm}$$

For fig. III

Applying following formula

$$\frac{\mu_1}{v} - \frac{\mu_2}{u} = \frac{\mu_2 - \mu_1}{R}$$

Here $\mu_2=1$, $\mu_1=1.5$, $u = -30$ cm, $v = -H_3$, $R= 3m = 300$ cm

$$\frac{1}{-H_3} - \frac{1.5}{-30} = \frac{1-1.5}{+300} = \frac{0.5}{300} = \frac{1}{600}$$

$$+\frac{1}{H_3} = \frac{1}{20} + \frac{1}{600}$$

$$\Rightarrow H_3 = \frac{600}{31} = 19.3$$

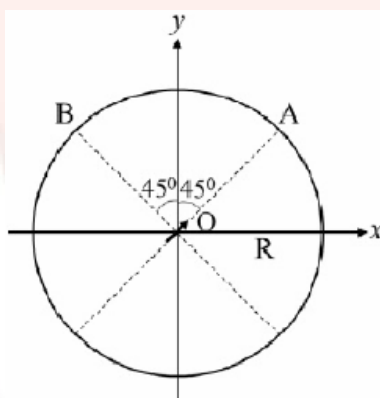
Now from above, we get, $H_1 = 20$ cm, $H_2 = 20.6$ cm, $H_3 = 19.3$ cm

$$H_2 - H_1 = 20.6 - 20 = 0.6$$

With above values, we can conclude that

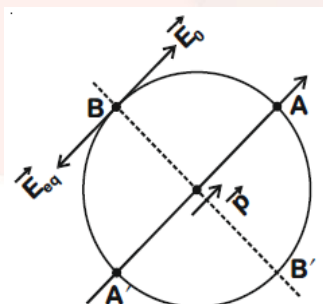
$$H_2 > H_1, H_3 < H_1, H_2 > H_3$$

2. An electric dipole with dipole moment $\frac{p_0}{\sqrt{2}}(\hat{i} + \hat{j})$ is held fixed at the origin O in the presence of an uniform electric field of magnitude E_0 . If the potential is constant on a circle of radius R centered at the origin as shown in figure, then the correct statement (s) is/are : (ϵ_0 is permittivity of free space $R \gg$ dipole size)



- (1) Total electric field at point B is $\vec{E}_B = 0$
- (2) Total electric field at point A is $\vec{E}_A = \sqrt{2}E_0(\hat{i} + \hat{j})$
- (3) $R = \left(\frac{p_0}{4\pi\epsilon_0 E_0} \right)^{1/3}$
- (4) The magnitude of total electric field on any two points of the circle will be same.

Soln.: [1, 3]



At point B, E due to dipole is equatorial. $E_{\text{equatorial}}$ at all points on circle should be zero since the given circle is equipotential

$$E_B = 0 \Rightarrow \frac{kp_0}{R^3} = E_0$$

So, option (1) is correct.

Electric field at point A, $E_A = \frac{2kp_0}{R^3} + E_0 = 2E_0 + E_0 = 3E_0$

$$E_0 = \frac{kp_0}{R^3} \Rightarrow R = \left(\frac{kp}{E_0} \right)^{1/3} = \left(\frac{p_0}{4\pi\epsilon_0 E_0} \right)^{1/3}$$

So, option (3) is correct.

Because E_0 is uniform & due to dipole Electric field is different at different points, so magnitude of total electric field will also be different at different points.

So, option (4) is incorrect.

3. A thin and uniform rod of mass M and length L is held vertical on a floor with large friction. The rod is released from rest so that it falls by rotating about its contact-point with the floor without slipping. Which of the following statement (s) is/are correct, when the rod makes an angle 60° with vertical? [g is the acceleration due to gravity]
- (1) The angular acceleration of the rod will be $\frac{2g}{L}$
 - (2) The radial acceleration of the rod's center of mass will be $\frac{3g}{4}$
 - (3) The angular speed of the rod will be $\sqrt{\frac{3g}{2L}}$

(4) The normal reaction force from the floor on the rod will be $\frac{Mg}{16}$

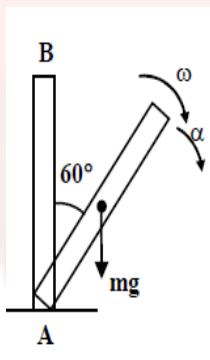
Soln.: [2, 3, 4]

By applying conservation of energy, we get

$$\frac{1}{2} I \omega^2 = mg \frac{L}{4}$$

$$\frac{1}{2} \frac{mL^2}{3} \omega^2 = mg \frac{L}{4}$$

$$\omega = \sqrt{\frac{3g}{2L}}$$



Now,

$$\tau = I\alpha \Rightarrow (mg) \frac{L}{2} \sin 60^\circ = \frac{ML^2}{3} \alpha$$

$$\alpha = \frac{3g}{2L} \sin 60^\circ = \frac{3g}{2L} \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}g}{4L}$$

$$\text{Tangential acceleration, } a_T = \frac{L}{2} \alpha = \frac{L}{2} \frac{3\sqrt{3}g}{4L} = \frac{3\sqrt{3}g}{8}$$

$$\text{Radial acceleration } a_c = \frac{L}{2} \omega^2 = \frac{L}{2} \frac{3g}{2L} = \frac{3g}{4}$$

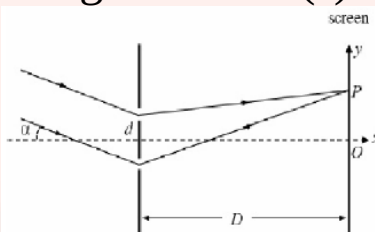
Net vertical acceleration of C.M. of rod

$$a_v = \frac{3g}{4} \cos 60^\circ + \frac{3\sqrt{3}g}{8} \sin 60^\circ = \frac{3g}{8} + \frac{3\sqrt{3}g}{8} \frac{\sqrt{3}}{2} = \frac{3g}{8} \left[1 + \frac{3}{2} \right] = \frac{15g}{16}$$

Applying $F_{\text{net}} = ma_v$ in vertical direction on rod as system

$$Mg - N = M \frac{15g}{16} \Rightarrow N = \frac{Mg}{16}$$

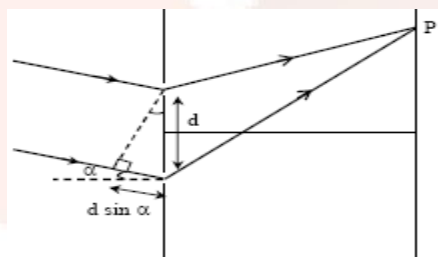
4. In a Young's double slit experiment, the slit separation d is 0.3 mm and the screen distance D is 1m. A parallel beam of light of wavelength 600 nm is incident on the slits at angle α as shown in figure. On the screen, the point O is equidistant from the slits and distance PO is 11.0 mm. which of the following statement(s) is/are correct?



- (1) For $\alpha = \frac{0.36}{\pi}$ degree, there will be destructive interference at point O
- (2) For $\alpha = 0$, there will be constructive interference at point P
- (3) For $\alpha = \frac{0.36}{\pi}$ degree, there will be destructive interference at point P .
- (4) Fringe spacing depends on α .

Soln.: [3]

Given $\alpha = \frac{0.36}{\pi} \text{ degree} = \frac{0.36}{\pi} \times \frac{\pi}{180} = \frac{1}{500} \text{ radian}$



$$\begin{aligned}
 \text{Path difference at } p &= d \sin \alpha + \frac{dy}{D} \quad (\because \alpha \rightarrow \text{small, } \sin \alpha = \alpha) \\
 &= d\alpha + \frac{dy}{D} \\
 &= (0.3 \times 10^{-3}) \left[\alpha + \frac{11 \times 10^{-3}}{1} \right]
 \end{aligned}$$

$$= \frac{3 \times 10^{-4} \left[\frac{1}{500} + 11 \times 10^{-3} \right]}{600 \times 10^{-9}} = \frac{10^3}{2} [13 \times 10^{-3}] = \frac{13}{2}$$

$$\frac{\Delta}{\lambda} = \frac{13}{2} = 6.5 \text{ Destructive interference}$$

$$(2) \quad \alpha = 0, \frac{\Delta}{\lambda} = \frac{3 \times 10^{-4} \times 11 \times 10^{-3}}{6 \times 10^{-7}} = \frac{11}{2} = 5.5 \text{ (Destructive)}$$

Fringe width in all the above case remain unchanged. So, Fringe width does not depend on α .

5. A mixture of ideal gas containing 5 moles of monatomic gas and 1 mole of rigid diatomic gas is initially at pressure P_0 , volume V_0 , and temperature T_0 . If the gas mixture is adiabatically compressed to a volume $V_0/4$, then the correct statement(s) is/are, (Given $2^{1.2} = 2.3$; $2^{3.2} = 9.2$; ; R is gas constant)

- (1) The final pressure of the gas mixture after compression is in between $9P_0$ and $10P_0$.
- (2) The work $|W|$ done during the process is $13 RT_0$.
- (3) Adiabatic constant of the gas mixture is 1.6.
- (4) The average kinetic energy of the gas mixture after compression is in between $18RT_0$ and $19RT_0$.

Soln.: [1, 2, 3]

Given:

$n_1 = 5$ moles of monatomic gas and $n_2 = 1$ mole of diatomic gas

Adiabatic constant of the gas mixture, $\gamma_{\text{mix}} = \frac{n_1 C_{p1} + n_2 C_{p2}}{n_1 C_{v1} + n_2 C_{v2}}$

$$C_{v1} = \frac{3R}{2} \quad C_{p1} = \frac{5R}{2}$$

$$C_{v2} = \frac{5R}{2} \quad C_{p2} = \frac{7R}{2}$$

$$\text{Adiabatic constant of the gas mixture} = \frac{5\left(\frac{5R}{2}\right) + (1)\left(\frac{7R}{2}\right)}{5\left(\frac{3R}{2}\right) + (1)\left(\frac{5R}{2}\right)} = \frac{32R/2}{20R/2} = \frac{8}{5} = 1.6$$

And for adiabatic process $P_0 V_0^\gamma = P \left(\frac{V_0}{4}\right)^\gamma$ ($\because$ adiabatically compression $= \frac{V_0}{4}$)

The final pressure of the gas, $P = P_0 (4)^\gamma$ $P = P_0 (4)^{8/5} = 9.2 P_0$

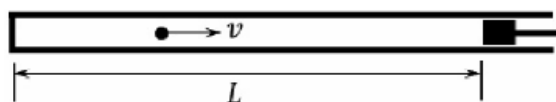
$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{P_0 V_0 - (4)^{8/5} P_0 \frac{V_0}{4}}{\frac{8}{5} - 1}$$

$$= P_0 V_0 \left[\frac{1 - 4^{3/5}}{3/5} \right] = P_0 V_0 \frac{-1.92}{0.6} = -2.15 P_0 V_0$$

$$|W| = 2.15(6)RT_0 = 13RT_0$$

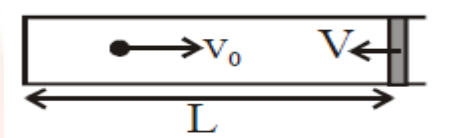
$$\text{Average kinetic energy} = nCVT = 6 \times \frac{5R}{3} \times 2.3 T_0 = 23 RT_0$$

6. A small particle of mass m moving inside a heavy, hollow and straight tube along the tube axis undergoes elastic collision at two ends. The tube has no friction and it is closed at one end by a flat surface while the other end is fitted with a heavy movable flat piston as shown in figure. When the distance of the piston from closed end is $L = L_0$ the particle speed is $v = v_0$. The piston is moved inward at a very low speed V and such that $v \ll \frac{dL}{L} v_0$, where dL is the infinitesimal displacement of the piston. Which of the following statement (s) is/are correct?



- (1) The particle's kinetic energy increases by a factor of 4 when the piston is moved inward from L_0 to $\frac{1}{2}L_0$.
- (2) After each collision with the piston, the particle speed increases by $2V$.
- (3) The rate at which the particle strikes the piston is v/L .
- (4) If the piston moves inward by dL , the particle speed increases by $2v \frac{dL}{L}$.

Soln.: [1, 2]



(3) The rate at which the particle strikes the piston = $\frac{2L}{v}$

(2) Speed of particle after collision = $2V + v_0$

Change in speed = $(2V + v_0) - v_0$

After each collision = $2V$

$$\text{Frequency} = \frac{v}{2L}$$

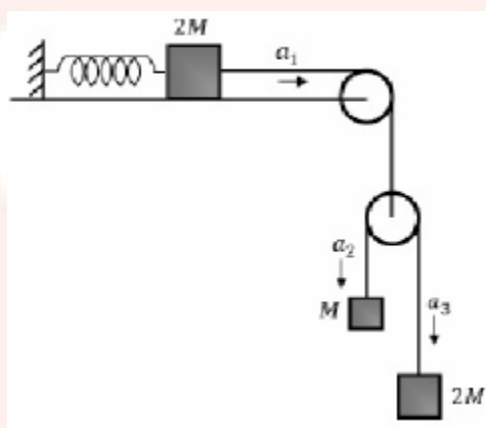
Change in speed in dt time = $2V \times \text{number of collision in } dt \text{ time}$

$$\Rightarrow dv = 2v \times \frac{v}{2L} \frac{dL}{V} = \frac{vdL}{L}$$

On decreasing length to half K.E. becomes $\frac{1}{4}$

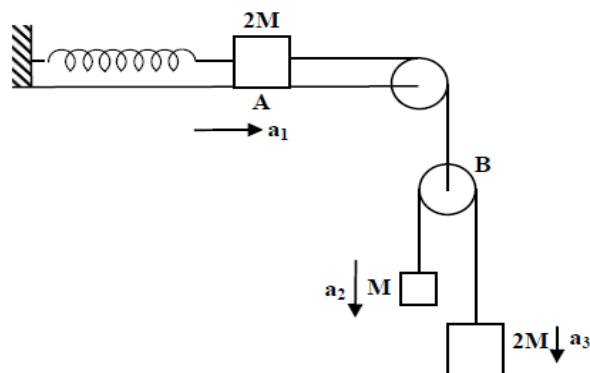
Option (1) and (2)

7. A block of mass $2M$ is attached to a massless spring with spring-constant k . This block is connected to two other blocks of masses M and $2M$ using two massless pulleys and strings. The accelerations of the blocks are a_1, a_2 and a_3 as shown in the figure. The system is released from rest with the spring in its unstretched state. The maximum extension of the spring is x_0 . Which of the following option (s) is/are correct? [g is the acceleration due to gravity. Neglect friction]



- (1) When spring achieves an extension of $\frac{x_0}{2}$ for the first time, the speed of the block connected to the spring is $3g\sqrt{\frac{M}{5k}}$
- (2) $x_0 = \frac{4Mg}{k}$
- (3) At an extension of $\frac{x_0}{4}$ of the spring, the magnitude of acceleration of the block connected to the spring is $\frac{3g}{10}$
- (4) $a_2 - a_1 = a_1 - a_3$

Soln.: [4]



$$2a_1 = a_2 + a_3$$

$$a_1 - a_3 = a_2 - a_1$$

$$a_p = \frac{a_2 + a_3}{2} \text{ or } a_2 - a_1 = a_1 - a_3$$

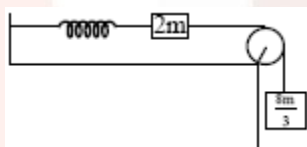
Also

$$\text{Reduced mass} = \frac{m \times 2m}{m + 2m} = \frac{2m}{3}$$

$$\text{Tension} = \frac{2m_1 m_2}{m_1 + m_2} g_{\text{eff}} = \frac{4m}{3} g_{\text{eff}}$$

$$2T = 2 \times \frac{4m}{3} g_{\text{eff}}$$

$$\frac{2T}{g_{\text{eff}}} = \frac{8m}{3}$$



$$\frac{8m}{3} \cdot x_0 g + = \frac{1}{2} k x_0^2$$

$$x_0 = \frac{16mg}{3k}$$

Energy conservation

$$\frac{8m}{3} g \frac{x_0}{2} = \frac{1}{2} k \frac{x_0^2}{4} + \frac{1}{2} (2m) v^2$$

$$\frac{16mg}{3k} \frac{4mg}{3} - \frac{1}{8} k \left(\frac{16mg}{3k} \right)^2 = mv^2 + \frac{4m}{3} v^2$$

$$\frac{32}{9} \frac{m^2 g^2}{k} = \frac{7}{3} m v^2$$

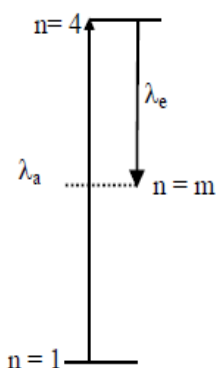
$$v^2 = \frac{32 \text{ mg}^2}{21 \text{ k}}$$

$$v = g \sqrt{\frac{32 \text{ mg}}{21 \text{ k}}}$$

8. A free hydrogen atom after absorbing a photon of wavelength λ_a gets excited from the state $n = 1$ to the state $n = 4$. Immediately after that the electron jumps to $n = m$ state by emitting a photon of wavelength λ_e . Let the change in momentum of atom due to the adsorption and the emission are Δp_a and Δp_e , respectively. If $\frac{\lambda_a}{\lambda_e} = \frac{1}{5}$, which of the option (s) is/are correct ? [Use $hc = 1242 \text{ eVnm}$; $1\text{nm} = 10^{-9}\text{m}$, h and c are Planck's constant and speed of light, respectively]
- (1) $\lambda_e = 418\text{nm}$
 - (2) The ratio of kinetic energy of the electron in the state $n = m$ to the state $n = 1$ is $\frac{1}{4}$
 - (3) $m = 2$
 - (4) $\frac{\Delta p_a}{\Delta p_e} = \frac{1}{2}$

Soln.: [2, 3]

The energy level diagram is shown in the figure



It is given that: $\frac{\lambda_a}{\lambda_e} = \frac{\frac{1}{m^2} - \left(\frac{1}{4}\right)^2}{1 - \left(\frac{1}{4}\right)^2} = \frac{1}{5}$

$$\frac{1}{m^2} - \frac{1}{16} = \frac{1}{5} \times \frac{15}{16} = \frac{3}{16}$$

$$\frac{1}{m^2} = \frac{4}{16} \Rightarrow m = 2$$

Now,

$$\frac{hc}{\lambda_e} = 13.6 \left(\frac{1}{4} - \frac{1}{16} \right)$$

$$\frac{1240}{\lambda_e} = 13.6 \left(\frac{12}{16 \times 4} \right)$$

$$\lambda_e = \frac{1240 \times 16 \times 4}{12 \times 13.6} = 487 \text{ nm}$$

The ratio of the kinetic energies is also the ratio of the corresponding total energies $= \frac{1}{4}$

$$\Delta p_a = \frac{h}{\lambda_a} \text{ and } \Delta p_e = \frac{h}{\lambda_e}$$

$$\text{Now, } \frac{\Delta p_a}{\Delta p_e} = \frac{\lambda_e}{\lambda_a} = 5$$

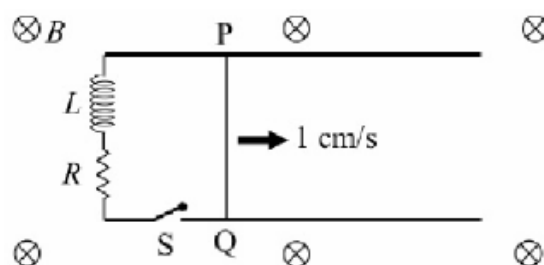
SECTION – 2 (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme.

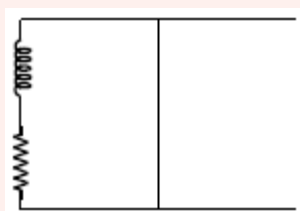
Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks:0 In all other cases.

1. A 10 cm long perfectly conducting wire PQ is moving with a velocity 1 cm/s on a pair of horizontal rails of zero resistance. One side of the rails is connected to an inductor $L = 1 \text{ mH}$ and a resistance $R = 1\Omega$ as shown in figure. The horizontal rails, L and R lie in the same plane with a uniform magnetic field $B = 1\text{T}$ perpendicular to the plan. If the key S is closed at certain instant, the current in the circuit after 1 millisecond is $x \times 10^{-3} \text{ A}$, where the value of x is
[Assume the velocity of wire PQ remains constant (1 cm/s) after key S is closed. Given: $e^{-1} = 0.37$, where e is base of the natural logarithm]



Soln.: [0.63]



Given: $B = 1\text{ T}$, $l = 10\text{ cm}$, $v = 1\text{ cm/s}$

Since velocity of PQ is constant. So emf developed across it remains constant

$E = vB$ where $l =$ length of wire PQ

$$= (10^{-2}\text{ m/s}) \left(\frac{10}{100} \right) (1) = 10^{-3}\text{ volt}$$

Current at any time t is given by

$$i = i_0(1 - e^{-t/\tau}) \quad \tau = \frac{L}{R} = \frac{10^{-3}}{1}$$

$$= \frac{10^{-3}}{1} \left[1 - e^{-10^{-3}/10^{-3}} \right]$$

$$= 10^{-3} \left(1 - \frac{1}{e} \right) = 0.63 \times 10^{-3}$$

Therefore,

$$x = 0.63$$

2. Suppose a ${}^{226}_{88}\text{Ra}$ nucleus at rest and in ground state undergoes α -decay to a ${}^{222}_{86}\text{Rn}$ nucleus in its excited state. The kinetic energy of the emitted α particle is found to be 4.44 MeV. ${}^{222}_{86}\text{Rn}$ nucleus then goes to its ground state by γ -decay. The energy of the emitted γ photon is keV. [Given: atomic mass of ${}^{226}_{88}\text{Ra} = 226.005\text{u}$, atomic mass of ${}^{222}_{86}\text{Rn} = 222.000\text{u}$, atomic mass of α particle = 4.000 u, $1\text{u} = 931\text{MeV}/c^2$, c is speed of the light]

Soln.: [135]

$$\text{Mass defect } \Delta m = (226.005) - [222.000 + 4.000] \\ = 226.005 - 226.000 = 0.005\text{amu}$$

$$Q = \Delta m c^2 = 0.005 \times 931.5 = 4.6575 \text{ MeV} = E_0$$

Since momentum is conserved, kinetic energy is in inverse ratio of masses.

$$\vec{p}_{\text{Rn}} = -\vec{p}_{\text{He}} \Rightarrow \text{KE} \propto \frac{1}{m}$$

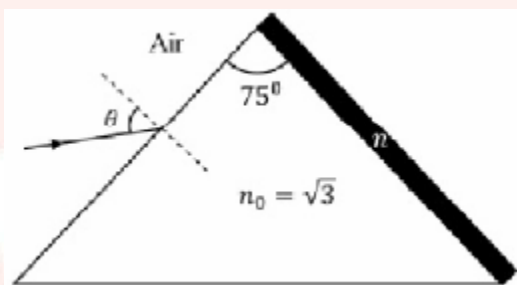
$$\frac{(\text{KE})_{\text{Rn}}}{(\text{KE})_{\alpha}} = \frac{m_{\alpha}}{m_{\text{Rn}}}$$

$$\Rightarrow (\text{KE})_{\text{Rn}} = \frac{4.000}{222.000} \times (\text{KE})_{\alpha}$$

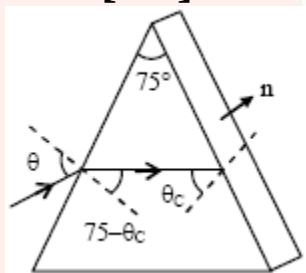
$$\Rightarrow (\text{KE})_{\text{Rn}} = \frac{4}{222} \times 4.44 \text{ MeV} = 0.08 \text{ MeV} \quad (\because \text{Given } (\text{KE})_{\alpha} = 4.44 \text{ MeV})$$

$$\text{Energy of } \gamma \text{ particle} = E_0 - \text{KE}_{\alpha} - \text{KE}_{\text{Rn}} = 4.65 - [4.44 + 0.08] \\ = 0.135 \text{ MeV} = 135 \text{ KeV}$$

3. A monochromatic light is incident from air on a refracting surface of a prism of angle 75° and refractive index $n_0 = \sqrt{3}$. The other refracting surface of the prism is coated by a thin film of material of refractive index n as shown in figure. The light suffers total internal reflection at the coated prism surface for an incidence angle of $\theta \leq 60^\circ$. The value of n^2 is



Soln.: [1.5]



Total internal reflection at the coated prism surface

$$\sin \theta_c = \frac{n}{\sqrt{3}} \dots \dots \dots (i)$$

By applying snell,s law

$$\sin \theta = \sqrt{3} \sin(75 - \theta_c)$$

For limiting condition $\theta = 60^\circ$,

$$\sin 60^\circ = \sqrt{3} \sin(75 - \theta_c) \Rightarrow \theta_c = 45^\circ$$

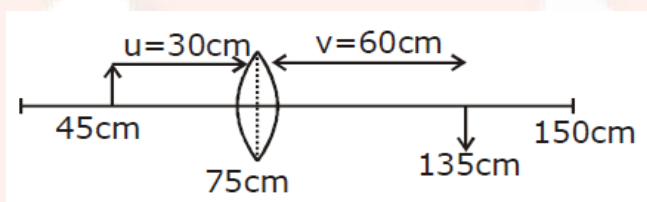
From equation (i), we get

$$\frac{1}{\sqrt{2}} = \frac{n}{\sqrt{3}} \Rightarrow n = \frac{\sqrt{3}}{\sqrt{2}} \Rightarrow n^2 = \frac{3}{2}$$

$$n^2 = 1.50$$

4. An optical bench has 1.5 m long scale having four equal divisions in each cm. While measuring the focal length of a convex lens, the lens is kept at 75 cm mark of the scale and the object pin is kept at 45 cm mark. The image of the object pin on the other side of the lens overlaps with image pin that is kept at 135 cm mark. In this experiment, the percentage error in the measurement of the focal length of the lens is

Soln.: [1.38 or 1.39]



For the given lens, $u = -30$ cm and $v = 60$ cm

By using lens formula, we get

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{60} - \frac{1}{-30} = \frac{1}{f}$$

$$f = 20 \text{ cm}$$

Now,

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

After differentiation both side, we get

$$\frac{df}{f^2} = \frac{dv}{v^2} + \frac{du}{u^2}$$

$$df = f^2 \left(\frac{dv}{v^2} + \frac{du}{u^2} \right) \dots\dots\dots(i)$$

$$\text{Here } f = 20 \text{ cm, } dv = du = 0.25 + 0.25 = 0.5 \text{ cm}$$

Putting values in above equation, we get

$$df = 400 \left(\frac{0.5}{(60)^2} + \frac{0.5}{(30)^2} \right) = \left(\frac{1}{36} + \frac{1}{9} \right) \times 2 = \frac{10}{36}$$

The percentage error in the measurement of the focal length of the lens is given by

$$\frac{df}{f} \times 100 = \frac{10/36}{20} \times 100 = \frac{50}{36} = 1.388$$

5. A ball is thrown from ground at an angle θ with horizontal and with an initial speed u_0 . For the resulting projectile motion, the magnitude of average velocity of the ball up to the point when it hits the ground for the first time is v_1 . After hitting the ground, the ball rebounds at the same angle θ but with a reduced speed of u_0/α . Its motion continues for a long time as shown in figure. If the magnitude of average velocity of the ball for entire duration of motion is $0.8v_1$, the value of α is



Soln.: [4]

Let $\frac{2u_0 \sin \theta}{g} = T_0$ and $u_0 \cos \theta = v_1$

$$V_{\text{avg}} = \frac{\text{Total displacement}}{\text{Time taken}}$$

Here, total displacement = $v_1 t_1 + v_2 t_2 + v_3 t_3 \dots$

Total time Taken = $t_1 + t_2 + t_3 \dots$

Now, we get

$$2 \left[\frac{u_0 \cos \theta u_0 \sin \theta}{g} \right] + 2 \left[\frac{\frac{u_0}{\alpha} \cos \theta \frac{u_0}{\alpha} \sin \theta}{g} \right]$$

$$= \frac{2u_0 \sin \theta}{g} + \frac{2 \frac{u_0}{\alpha} \sin \theta}{g} + \dots$$

$$= \frac{u_0 \cos \theta \left[1 + \frac{1}{\alpha^2} + \frac{1}{\alpha^4} + \dots + \frac{1}{\alpha^{2m}} \right]}{1 + \frac{1}{\alpha} + \frac{1}{\alpha^2} + \dots + \frac{1}{\alpha^m}}$$

$$m \rightarrow \infty$$

$$= V_1 \left[\frac{1 + \frac{1}{\alpha^2} + \frac{1}{\alpha^4} + \dots}{1 + \frac{1}{\alpha} + \frac{1}{\alpha^2} + \dots} \right]$$

$$= V_1 \left[\frac{\frac{1}{1 - \frac{1}{\alpha^2}}}{\frac{1}{1 - \frac{1}{\alpha}}} \right] = v_1 \left[\frac{\frac{\alpha^2}{\alpha^2 - 1}}{\frac{\alpha}{\alpha - 1}} \right]$$

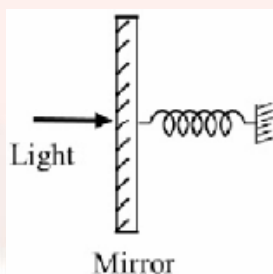
$$V_{\text{avg}} = V_1 \left[\frac{\alpha^2}{\alpha^2 - 1} \frac{\alpha - 1}{\alpha} \right] = \frac{\alpha}{\alpha + 1} V_1$$

$$0.8V_1 = V_1 = \frac{\alpha}{\alpha + 1} \Rightarrow \alpha = 0.8\alpha + 0.8$$

$$0.2\alpha = 0.8$$

$$\alpha = 4$$

6. A perfectly reflecting mirror of mass M mounted on a spring constitutes a spring-mass system of angular frequency Ω such that $\frac{4\pi M\Omega}{h} = 10^{24} \text{ m}^{-2}$ with h as Planck's constant. N photons of wavelength $\lambda = 8\pi \times 10^{-6} \text{ m}$ strike the mirror simultaneously at normal incidence such that the mirror gets displaced by $1 \mu\text{m}$. If the value of N is $x \times 10^{12}$, then the value of x is [Consider the spring as massless]



Soln.: [1]

Given that:

$$A = 1 \mu\text{m} = 10^{-6} \text{ m} \quad N = \text{No. of photons}, \quad \lambda = 8\pi \times 10^{-6} \text{ m}$$

If v is speed of mirror just after absorption of photons

$$Mv_{\max} = 2N \frac{h}{\lambda}$$

$$Ma\omega = 2N \frac{h}{\lambda}$$

$$M(10^{-6})(\Omega) = 2N \frac{h}{8\pi \times 10^{-6}}$$

$$10^{-12} = \frac{Nh}{4\pi M\Omega}$$

$$N = \left(\frac{4\pi M\Omega}{h} \right) 10^{-12}$$

$$N = 10^{24} \times 10^{-12} = 10^{12}$$

$$\text{(Given that: } \frac{4\pi M\Omega}{h} = 10^{24} \text{ m}^{-2} \text{)}$$

$$X = 1$$

SECTION – 3 (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **TWO (02)** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and List-II has Six entries (P), (Q), (R), (S), (T) and (U).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- For each question, marks will be awarded according to the following marking scheme :
Full Marks: +3 If **ONLY** the option corresponding to the correct matching is chosen.
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered.)
Negative Marks: –1 In all other cases.

Answer the following by appropriately matching the lists based on the information given in the paragraph.

1. A musical instrument is made using four different metal strings, 1, 2, 3 and 4 with mass per unit length $\mu, 2\mu, 3\mu$ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0 .

List-I gives the above four strings while List-II lists the magnitude of some quantity.

List - I		List – II	
(I)	String – 1 (μ)	(P)	1
(II)	String – 2 (2μ)	(Q)	$1/2$
(III)	String – 3 (3μ)	(R)	$1/\sqrt{2}$
(IV)	String – 4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	$3/16$
		(U)	$1/16$

If the tension in each string is T_0 , the correct match for the highest fundamental frequency in f_0 units will be,

- (1) I $\rightarrow$ Q; II $\rightarrow$ S; III $\rightarrow$ R; IV $\rightarrow$ P
- (2) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ S; IV $\rightarrow$ Q
- (3) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ S
- (4) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ R; IV $\rightarrow$ T

Soln.: [2]

For fundamental mode

$$\text{Frequency } f_0 = \frac{1}{2L_0} \sqrt{\frac{T}{\mu}}$$

$$\text{String (1)} \quad f_0 = \frac{1}{2L_0} \sqrt{\frac{T_0}{\mu}} \rightarrow P$$

$$\text{String (2)} \quad f_2 = \frac{1}{2L_0} \sqrt{\frac{T_0}{2\mu}} = \frac{f_0}{\sqrt{2}} \rightarrow R$$

String (3) $f_3 = \frac{1}{2L_0} \sqrt{\frac{T}{3\mu}} \rightarrow S$

String (4) $f_4 = \frac{1}{2L_0} \sqrt{\frac{T}{4\mu}} = \frac{f_0}{2} \rightarrow Q$

$I \rightarrow P, II \rightarrow R, III \rightarrow S, IV \rightarrow Q$ Option (2)

2. Answer the following by appropriately matching the lists based on the information given in the paragraph.

A musical instrument is made using four different metal strings, 1, 2, 3 and 4 with mass per unit length $\mu, 2\mu, 3\mu$ and 4μ respectively. The instrument is played by vibrating the strings by varying the free length in between the range L_0 and $2L_0$. It is found that in string-1 (μ) at free length L_0 and tension T_0 the fundamental mode frequency is f_0 .

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(IV)	String – 4 (4μ)	(S)	$1/\sqrt{3}$
		(T)	$3/16$
		(U)	$1/16$

The length of strings 1, 2, 3 and 4 are kept fixed at L_0 , $\frac{3L_0}{2}$, $\frac{5L_0}{4}$ and $\frac{7L_0}{4}$, respectively. String 1, 2, 3 and 4 are vibrated at their 1st, 3rd, 5th, and 14th harmonics, respectively such that all the strings have same frequency. The correct match for the tension in the four strings in the units of T_0 will be.

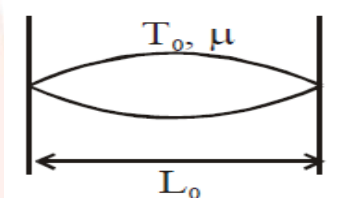
- (1) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ T; IV $\rightarrow$ U
- (2) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ T
- (3) I $\rightarrow$ T; II $\rightarrow$ Q; III $\rightarrow$ R; IV $\rightarrow$ U
- (4) I $\rightarrow$ P; II $\rightarrow$ R; III $\rightarrow$ T; IV $\rightarrow$ U

Soln.: [1]

For string (1)

Length of string = L_0

It is vibrating in 1st harmonic i.e. fundamental mode.

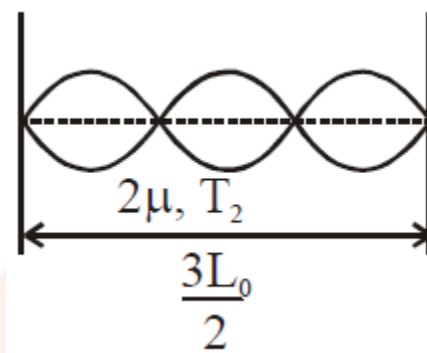


$$f_0 = \frac{1}{2L_0} \sqrt{\frac{T}{\mu}}$$

For string (2),

Length of string = $\frac{3L_0}{2}$

It is vibrating in 3rd harmonic but frequency is f_0



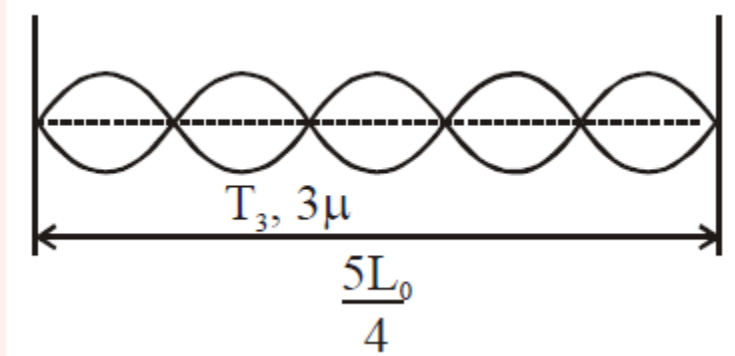
$$f_0 = \frac{3}{2 \left[\frac{3L_0}{2} \right]} \sqrt{\frac{T_2}{2\mu}} = \frac{1}{\sqrt{2}} \frac{1}{L_0} = \frac{1}{2L_0} \sqrt{\frac{T_0}{\mu}}$$

$$T_2 = \frac{T_0}{2}$$

For string (3)

Length of string = $\frac{5L_0}{4}$

It is vibrating in 5th harmonic but frequency is still f_0



$$f_0 = \frac{5}{2 \left[\frac{5L_0}{4} \right]} \sqrt{\frac{T_3}{3\mu}} = \frac{2}{\sqrt{3}} \frac{1}{L_0} \sqrt{\frac{T_3}{\mu}} = \frac{1}{2L_0} \sqrt{\frac{T_0}{\mu}}$$

$$T_3 = \frac{3}{16} T_0$$

For string (4),

Length of string = $\frac{7L_0}{4}$

It is vibrating in 5th harmonic but frequency is still f_0

$$f_0 = \frac{14}{2 \left[\frac{7L_0}{4} \right]} \sqrt{\frac{T_4}{4\mu}} = \frac{1}{2L_0} \sqrt{\frac{T_0}{\mu}}$$

$$\sqrt{4T_4} = \frac{\sqrt{T_0}}{2} \Rightarrow T_4 = \frac{T_0}{16}$$

3. Answer the following by appropriately matching the lists based on the information given in the paragraph.

In a thermodynamic process on an ideal monoatomic gas, the infinitesimal heat absorbed by the gas is given by $T_{\Delta X}$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas

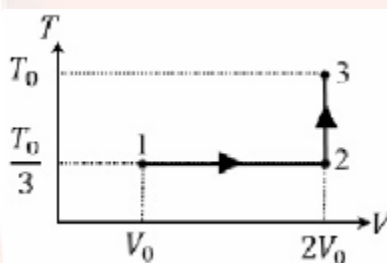
$X = R \ln \left(\frac{T}{T_A} \right) + R \ln \left(\frac{V}{V_A} \right)$. Here, R is gas constant, V is volume of gas, T_A and V_A are constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List – I		List – II	
(I)	Work done by the system in process $1 \rightarrow 2 \rightarrow 3$	(P)	$\frac{1}{3}RT_0 \ln 2$
(II)	Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$	(R)	RT_0

(IV)	Heat absorbed by the system in process $1 \rightarrow 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3 + \ln 2)$
		(U)	$\frac{5}{6}RT_0$

If the process on one mole of monatomic ideal gas is as shown in the TV-diagram with $p_0 V_0 = \frac{1}{3}RT_0$, the correct match is.



- (1) $I \rightarrow P$; $II \rightarrow R$; $III \rightarrow T$; $IV \rightarrow P$
- (2) $I \rightarrow P$; $II \rightarrow T$; $III \rightarrow Q$; $IV \rightarrow T$
- (3) $I \rightarrow P$; $II \rightarrow R$; $III \rightarrow T$; $IV \rightarrow S$
- (4) $I \rightarrow S$; $II \rightarrow T$; $III \rightarrow Q$; $IV \rightarrow U$

Soln.: [1]

Given that Process $1 \rightarrow 2$ is isothermal since temperature is constant.

Process $2 \rightarrow 3$ is isochoric since volume is constant

(1) Work done by the system in process $1 \rightarrow 2 \rightarrow 3$

$$W_{1-2-3} = W_{1 \rightarrow 2} + W_{2 \rightarrow 3}$$

$$W_{1-2-3} = nRT \ln \left(\frac{2V_0}{V_0} \right) + 0$$

$$W_{1-2-3} = nR \frac{T_0}{3} \ln 2 + 0 = \frac{RT_0}{3} \ln 2$$

(2) Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$

$$\begin{aligned}\Delta U_{123} &= \Delta U_{12} + \Delta U_{23} \\ &= 0 + nC_v dT \\ &= 0 + 1 \times \frac{3R}{2} \left(T_0 - \frac{T_0}{3} \right) = RT_0\end{aligned}$$

(3) Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$

$$\begin{aligned}\Delta Q_{123} &= \Delta Q_{12} + \Delta Q_{23} \\ &= \frac{RT_0}{3} \ln 2 + nC_v dT \\ &= \frac{RT_0}{3} \ln 2 + RT_0 \\ &= \frac{RT_0}{3} (\ln 2 + 3)\end{aligned}$$

(4) Heat absorbed by the system in process $1 \rightarrow 2$ (isothermal)

$$\begin{aligned}\Delta Q_{12} &= \Delta W \\ &= \frac{RT_0}{3} \ln 2\end{aligned}$$

4. **Answer the following by appropriately matching the lists based on the information given in the paragraph.**

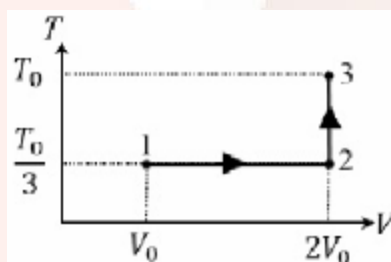
In a thermodynamic process on an ideal monoatomic gas, the infinitesimal heat absorbed by the gas is given by $T \Delta X$, where T is temperature of the system and ΔX is the infinitesimal change in a thermodynamic quantity X of the system. For a mole of monatomic ideal gas

$X = R \ln \left(\frac{T}{T_A} \right) + R \ln \left(\frac{V}{V_A} \right)$. Here, R is gas constant, V is volume of gas, T_A and V_A are constants.

The List-I below gives some quantities involved in a process and List-II gives some possible values of these quantities.

List – I		List – II	
(I)	Work done by the system in process $1 \rightarrow 2 \rightarrow 3$	(P)	$\frac{1}{3}RT_0 \ln 2$
(II)	Change in internal energy in process $1 \rightarrow 2 \rightarrow 3$	(Q)	$\frac{1}{3}RT_0$
(III)	Heat absorbed by the system in process $1 \rightarrow 2 \rightarrow 3$	(R)	RT_0
(IV)	Heat absorbed by the system in process $1 \rightarrow 2$	(S)	$\frac{4}{3}RT_0$
		(T)	$\frac{1}{3}RT_0(3 + \ln 2)$
		(U)	$\frac{5}{6}RT_0$

If the process on one mole of monatomic ideal gas is as shown in the TV-diagram with $p_0V_0 = \frac{1}{3}RT_0$, the correct match is.



- (1) I → Q; II → R; III → S; IV → U
- (2) I → Q; II → S; III → R; IV → U
- (3) I → S; II → R; III → Q; IV → T
- (4) I → Q; II → R; III → P; IV → U

Soln.: [1]

Given that Process 1 $\rightarrow$ 2 is isothermal since temperature is constant

Process 2 $\rightarrow$ 3 is isochoric since volume is constant

Degree of freedom $f = 3$

(1) Work done by the system in process 1 $\rightarrow$ 2 $\rightarrow$ 3

Work done in any process = Area under P–V graph

$$\begin{aligned} W_{123} &= w_{12} + w_{23} \\ &= P_0 V_0 = \frac{RT_0}{3} \end{aligned}$$

(2) Change in internal energy in process 1 $\rightarrow$ 2 $\rightarrow$ 3

$$\begin{aligned} \Delta U_{123} &= \Delta U_{12} + \Delta U_{23} \\ &= nC_v(T_3 - T_1) \\ &= n \frac{3}{2} R(T_3 - T_1) \\ &= \frac{3}{2} (RT_3 - RT_1) \\ &= \frac{3}{2} (3P_0 V_0 - P_0 V_0) = 3P_0 V_0 = RT_0 \end{aligned}$$

(3) Heat absorbed by the system in process 1 $\rightarrow$ 2 $\rightarrow$ 3

$$\begin{aligned} \Delta Q_{123} &= \Delta Q_{12} + \Delta Q_{23} \\ &= nC_p(T_2 - T_1) + nC_v(T_3 - T_2) \\ &= \frac{5}{2} (RT_2 - RT_1) + \frac{3}{2} (RT_3 - RT_2) \\ &= \frac{5}{2} (2P_0 V_0 - P_0 V_0) + \frac{3}{2} (3P_0 V_0 - 2P_0 V_0) \\ &= \frac{5}{2} P_0 V_0 + \frac{3}{2} P_0 V_0 = 4P_0 V_0 = \frac{4}{3} P_0 V_0 \end{aligned}$$

3 $\rightarrow$ S

(4) Heat absorbed by the system in process 1 $\rightarrow$ 2

$$\Delta Q_{12} = 5P_0 V_0 = \frac{5}{6} RT_0$$

4 $\rightarrow$ U

JEE Advanced Exam 2019
(Paper & Solution)
PAPER-2

PART-I (CHEMISTRY)

SECTION – 1(Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :
Full Marks: **+4** If only (all) the correct option(s) is (are) chosen.
Partial Marks: **+3** If all the four options are correct but **ONLY** three options are chosen.
Partial Marks: **+2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks: **+1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks: **0** If none of the option is chosen (i.e. the question is unanswered).
Negative Marks: **–1** In all other cases.

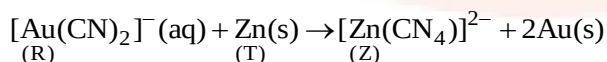
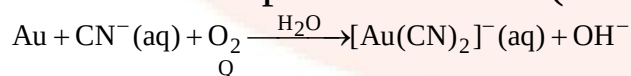
- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 marks; choosing ONLY (B) will get +1 marks; choosing ONLY (D) will get +1 marks; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get –1 mark

1. The cyanide process of gold extraction involves leaching out gold from its ore with CN^- in the presence of Q in water to form R. Subsequently, R is treated with T to obtain Au and Z. Choose the correct option(s)

- (1) Z is $[\text{Zn}(\text{CN})_4]^{2-}$ (2) T is Zn
(3) Q is O_2 (4) R is $[\text{Au}(\text{CN})_2]^-$

Soln.: [1, 2, 3]

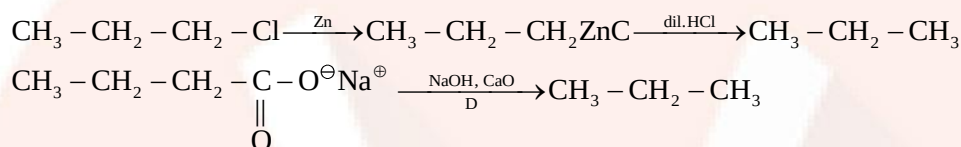
Extraction of gold is done by leaching with dil. solution of NaCN in the presence of air (CO_2).



2. Which of the following reactions produce(s) propane as a major product?

- (1) $\text{H}_3\text{C}-\text{CH}(\text{Br})-\text{CH}_2\text{Br} \xrightarrow{\text{Zn}}$
- (2) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2\text{Cl} \xrightarrow{\text{Zn, dil. HCl}}$
- (3) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2\text{COONa} + \text{H}_2\text{O} \xrightarrow{\text{electrolysis}}$
- (4) $\text{H}_3\text{C}-\text{CH}_2-\text{CH}_2\text{COONa} \xrightarrow{\text{NaOH, CaO, } \Delta}$

Soln.: [2, 4]



3. With reference to aqua regia, choose the correct option(s):

- (1) Reaction of gold with aqua regia produces NO_2 in the absence of air
- (2) Reaction of gold with aqua produces an anion having Au in +3 oxidation state
- (3) The yellow colour of aqua regia is due to the presence of NOCl and Cl_2
- (4) Aqua regia is prepared by mixing conc. HCl and conc. HNO_3 in 3:1 (v/v) ratio

Soln.: [2, 3, 4]

- (1) $2\text{Au} + 3\text{HNO}_3 + 11\text{HCl} \rightarrow 2\text{HAuCl}_4 + 3\text{NOCl} + 6\text{H}_2\text{O}$
- (2) In HAuCl_4 or $[\text{AuCl}_4]^-$, Au is in + 3 oxidation state.
- (3) Yellow colour of aqua regia is due to the presence of NOCl (orange yellow) and Cl_2 (greenish yellow)
- (4) Aqua regia is mixture of 3 HCl (conc.) and 1 HNO_3 .

4. The ground state energy of hydrogen atom is -13.6 eV. Consider an electronic state ψ of He^+ whose energy, azimuthal quantum number and magnetic quantum number are -3.4 eV, 2 and 0, respectively. Which of the following statement(s) is (are) true for the state ψ ?
- (1) The nuclear charge experienced by the electron in this state is less than $2e$, where e is the magnitude of the electronic charge
 - (2) It has 2 angular nodes
 - (3) It has 3 radial nodes
 - (4) It is a 4d state

Soln.: [2, 4]

$$\text{He}^+ = (z = 2)$$

Ground state energy is given

$$E_n = -13.6 \frac{z^2}{n^2} = -3.4$$

$$-13.6 \frac{(2)^2}{n^2} = -3.4$$

$$n^2 = 4^2$$

$$n = 4$$

Quantum number are

$$n = 4, l = 2, m = 0$$

So, subshell is 4d.

Angular node = $l = 2$

$$\text{Radial node} = n - l - 1 = 4 - 2 - 1 = 1$$

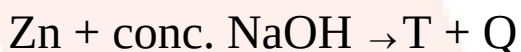
Nuclear charge = $2e$

5. Choose the correct option(s) from the following:
- (1) Cellulose has only α -D-glucose units that are joined by glycosidic linkages
 - (2) Natural rubber is polyisoprene containing trans alkene units
 - (3) Nylon-6 has amide linkages
 - (4) Teflon is prepared by heating tetrafluoroethene in presence of a persulphate catalyst at high

Soln.: [3, 4]

- (1) Cellulose has only β -D-glucose units that are joined by glycosidic linkages.
- (2) Natural Rubber units which are joined by $C_1 - C_4$ glycosidic linkage is cis-polyisoprene.
- (3) Nylon-6 has amide linkages
- (4) Teflon is prepared by heating polytetrafluoro ethene in presence of a persulphate catalyst.

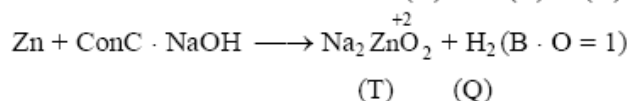
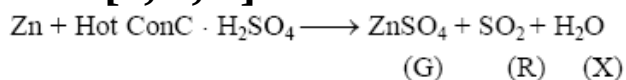
6. Consider the following reactions (unbalanced)

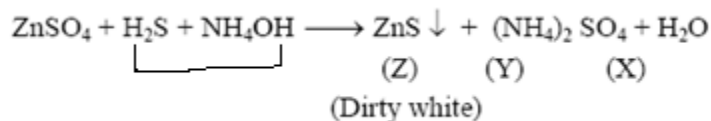


Choose the correct option(s)

- (1) Bond order of Q is 1 in its ground state
- (2) Z is dirty white in color
- (3) R is a V-shaped molecule
- (4) The oxidation state of Zn in T is +1

Soln.: [1, 2, 3]



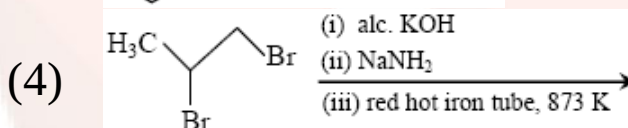
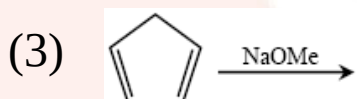
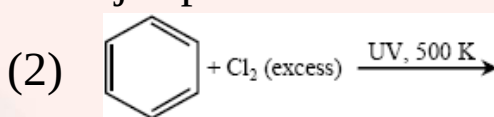
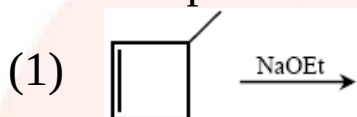


$\text{SO}_2(\text{R})$ is V-shaped

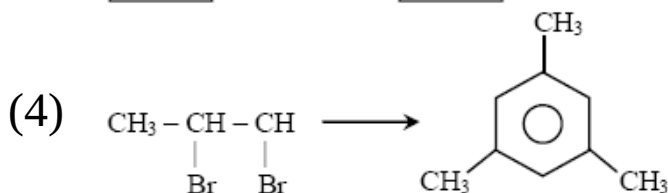
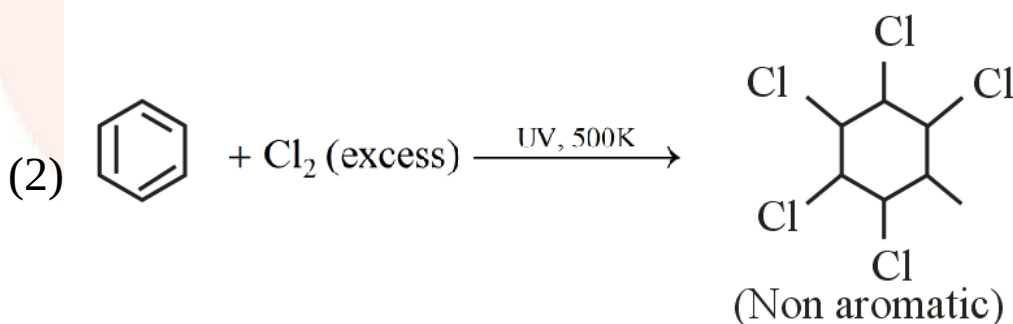
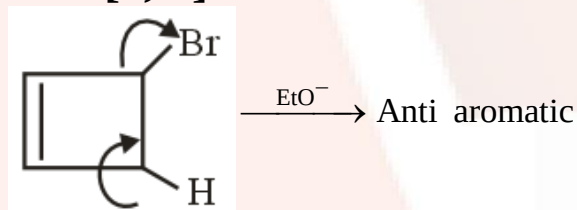
$\text{ZnS}(\text{z})$ is dirty white gelatinous ppt.

Bond order in $\text{H}_2(\theta)$ is 1.

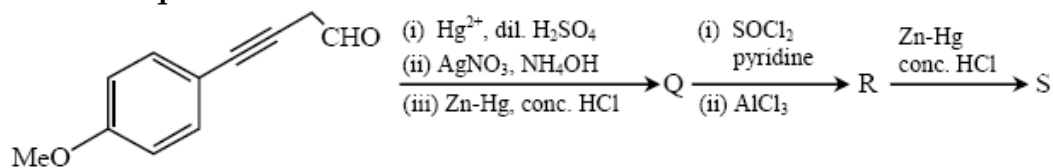
7. Choose the correct option(s) that give(s) an aromatic compound as the major product:



Soln.: [3, 4]



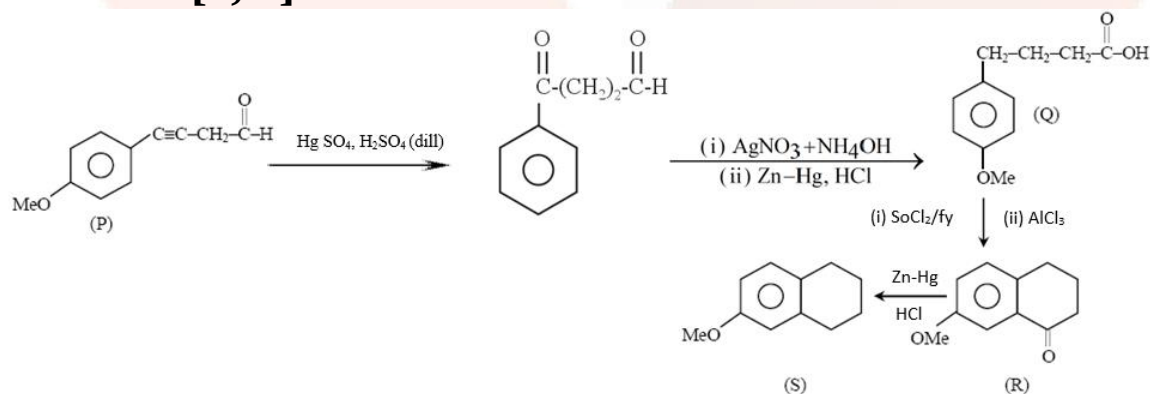
8. Choose the correct option(s) for the following reaction sequence



Consider Q, R and S as major products:

- (1)
- (2)
- (3)
- (4)
-
-
-

Soln.: [1, 4]



SECTION – 2 (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to TWO decimal places.
- Answer to each question will be evaluated according to the following marking scheme.

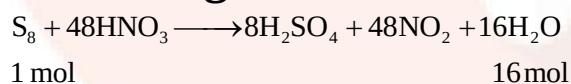
Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

1. The amount of water produced (in g) in the oxidation of 1 mole of rhombic sulphur by conc. HNO_3 to a compound with the highest oxidation state of sulphur is _____

(Given data: Molar mass of water = 18 g mol^{-1})

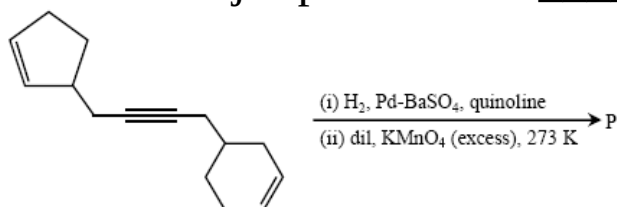
Soln.: [288 gm]



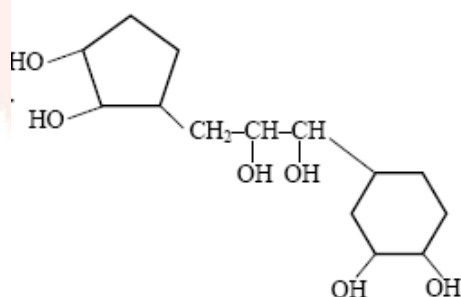
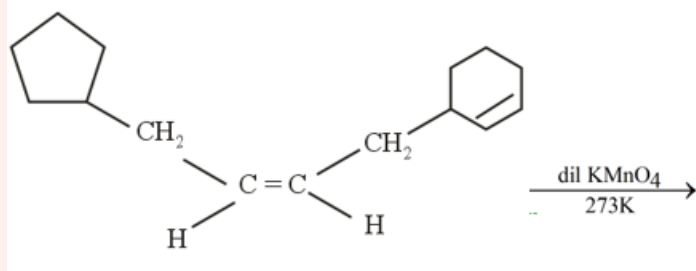
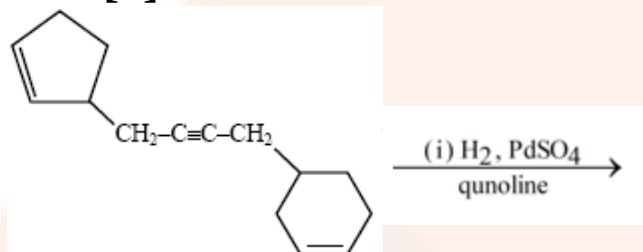
Mole of water = 16

Amount of water produced = $16 \times 18 = 288 \text{ gm}$

2. Total number of hydroxyl groups present in a molecule of the major product P is _____



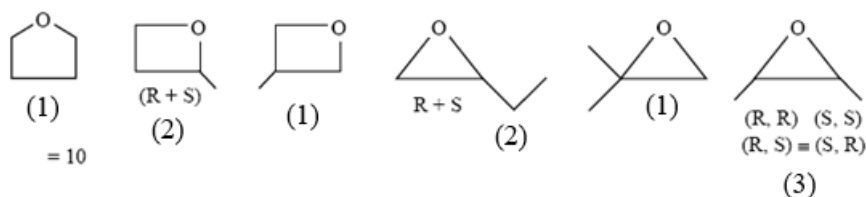
Soln.: [6]



Total 6-OH group present in a molecule of the product.

3. Total number of isomers, considering both structural and stereoisomers, of cyclic ethers with the molecular formula C_4H_8O is _____

Soln.: [10]



4. The mole fraction of urea in an aqueous urea solution containing 900 g of water is 0.05. If the density of the solution is 1.2 g cm^{-3} , the molarity of urea solution is _____

(Given data: Molar masses of urea and water are 60 g mol^{-1} and 18 g mol^{-1} , respectively)

Soln.: [2.98]
mole fraction urea

$$X_{\text{urea}} = \frac{n_{\text{urea}}}{n_{\text{urea}} + n_{\text{water}}} = 0.05$$

$$\frac{n_{\text{urea}}}{n_{\text{urea}} + \frac{900}{18}} = 0.05$$

$$\frac{n_{\text{urea}}}{n_{\text{urea}} + 50} = 0.05$$

$$n_{\text{urea}} = 0.05n_{\text{urea}} + 2.5$$

$$0.95n_{\text{urea}} = 2.5$$

$$n_{\text{urea}} = \frac{2.5}{0.95} = \frac{250}{95} = 2.63 \text{ mol}$$

$$\begin{aligned} \text{Mass of urea} &= 2.63 \times 60 \quad (\because \text{molar mass of urea} = 60 \text{ g/mol}) \\ &= 157.89 \end{aligned}$$

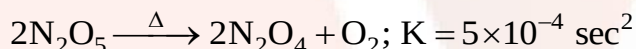
$$\text{Mass of solution} = 157.8 + 900 = 1057.8\text{g}$$

$$\text{Volume of solution} = \frac{1057.8}{1.2} = 881.5\text{ml}$$

$$\text{Molality of solution} = \frac{2.63}{881.5} \times 1000 = 2.98\text{M}$$

5. The decomposition reaction $2\text{N}_2\text{O}_5(\text{g}) \xrightarrow{\Delta} 2\text{N}_2\text{O}_4(\text{g}) + \text{O}_2(\text{g})$ is started in a closed cylinder under isothermal isochoric condition at an initial pressure of 1 atm. After $Y \times 10^3\text{s}$, the pressure inside the cylinder is found to be 1.45 atm. If the rate constant of the reaction is $5 \times 10^{-4}\text{s}^{-1}$, assuming ideal gas behaviour, the value of Y is _____

Soln.: [2.30]



Initial pressure (in atm) 1 - -

Pressure after time t sec $1 - 2x$ $2x$ x

$$\therefore 1 - 2x + 2x + x = 1.45 \text{ atm}$$

$$\therefore x = 0.45 \text{ atm}$$

For 1st order reaction (taking into account of reaction stoichiometry)

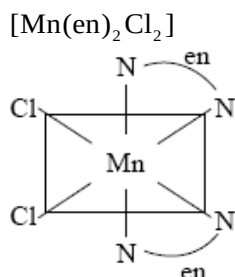
$$2Kt = \ln \frac{a_0}{a_0 - x}$$

$$\therefore a_0 \propto 1 \quad \text{and} \quad a_0 - x \propto 0.1$$

$$\Rightarrow t = \frac{1}{2 \times 5 \times 10^{-4}} \ln \frac{1}{0.1} = 2.3 \times 10^3 \text{ sec}$$

6. Total number of cis N–Mn–Cl bond angles (that is, Mn–N and Mn–Cl bonds in cis positions) present in a molecule of $\text{cis} - [\text{Mn}(\text{en})_2\text{Cl}_2]$ complex is _____ (en = $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$)

Soln.: [6]



Number of cis (Cl – Mn - N) = 6

SECTION – 3 (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **TWO (02)** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **four** entries (I), (II), (III) and (IV) and List-II has Six entries (P), (Q), (R), (S), (T) and (U).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- For each question, marks will be awarded according to the following marking scheme :
Full Marks: +3 If **ONLY** the option corresponding to the correct matching is chosen.

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered.)

Negative Marks: -1 In all other cases.

Answer the following by appropriately matching the lists based on the information given in the paragraph.

1. Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following. List-I contains some quantities for the n^{th} orbit of the atom and List-II contains options showing how they depend on n .

LIST-I		LIST-II	
(I)	Radius of the n^{th} orbit	(P)	$\propto n^{-2}$
(II)	Angular momentum of the electron in the n^{th} orbit	(Q)	$\propto n^{-1}$
(III)	Kinetic energy of the electron in the n^{th} orbit	(R)	$\propto n^0$
(IV)	Potential energy of the electron in the n^{th} orbit	(S)	$\propto n^1$
		(T)	$\propto n^2$
		(U)	$\propto n^{1/2}$

(1) (III), (P)

(2) (III), (S)

(3) (IV), (U)

(4) (IV), (Q)

Soln. [1]

(I) Radius of the nth orbit

$$r = 0.529 \frac{n^2}{Z}$$

$$r \propto n^2$$

$$I \rightarrow T$$

(II) Angular momentum of the electron in the nth orbit

$$= mvr = \frac{nh}{2\pi}$$

$$mvr \propto n$$

$$II \rightarrow S$$

(III) Kinetic energy of the electron in the nth orbit.

$$K.E = \frac{13.6 \times Z^2}{n^2} \text{ eV}$$

$$KE \propto n^{-2}$$

$$III \rightarrow P$$

(IV) Potential energy of the electron in the nth orbit.

$$P.E = -2.KE = -27.2 \times \frac{Z^2}{n^2} \text{ eV}$$

$$\Rightarrow PE \propto n^{-2}$$

$$\therefore (IV) \rightarrow (P)$$

2. Consider the Bohr's model of a one-electron atom where the electron moves around the nucleus. In the following. List-I contains some quantities for the n^{th} orbit of the atom and List-II contains options showing how they depend on n .

LIST-I		LIST-II	
(I)	Radius of the n^{th} orbit	(P)	$\propto n^{-2}$
(II)	Angular momentum of the electron in the n^{th} orbit	(Q)	$\propto n^{-1}$
(III)	Kinetic energy of the electron in the n^{th} orbit	(R)	$\propto n^0$
(IV)	Potential energy of the electron in the n^{th} orbit	(S)	$\propto n^1$
		(T)	$\propto n^2$
		(U)	$\propto n^{1/2}$

(1) (I), (P)

(2) (I), (T)

(3) (II), (R)

(4) (II), (Q)

Soln.: [2]

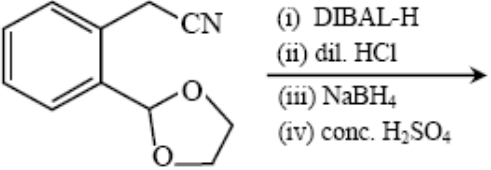
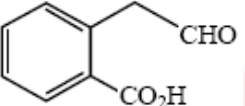
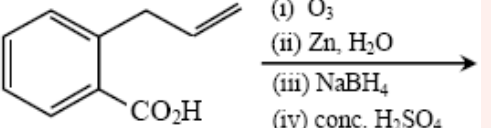
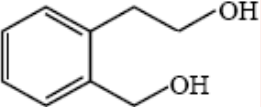
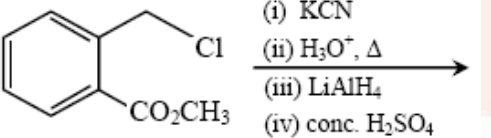
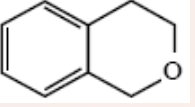
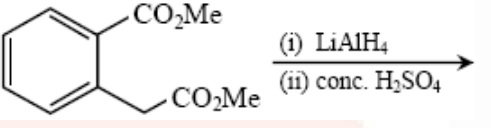
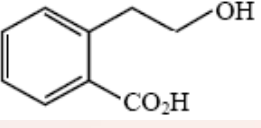
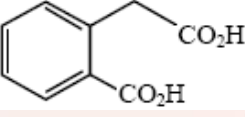
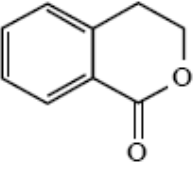
Radius $r \propto n^2$

Kinetic energy $\propto n^{-2}$

Potential energy $\propto n^{-2}$

$\therefore$ Correct match: (I), (T); (II), (S)

3. List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of

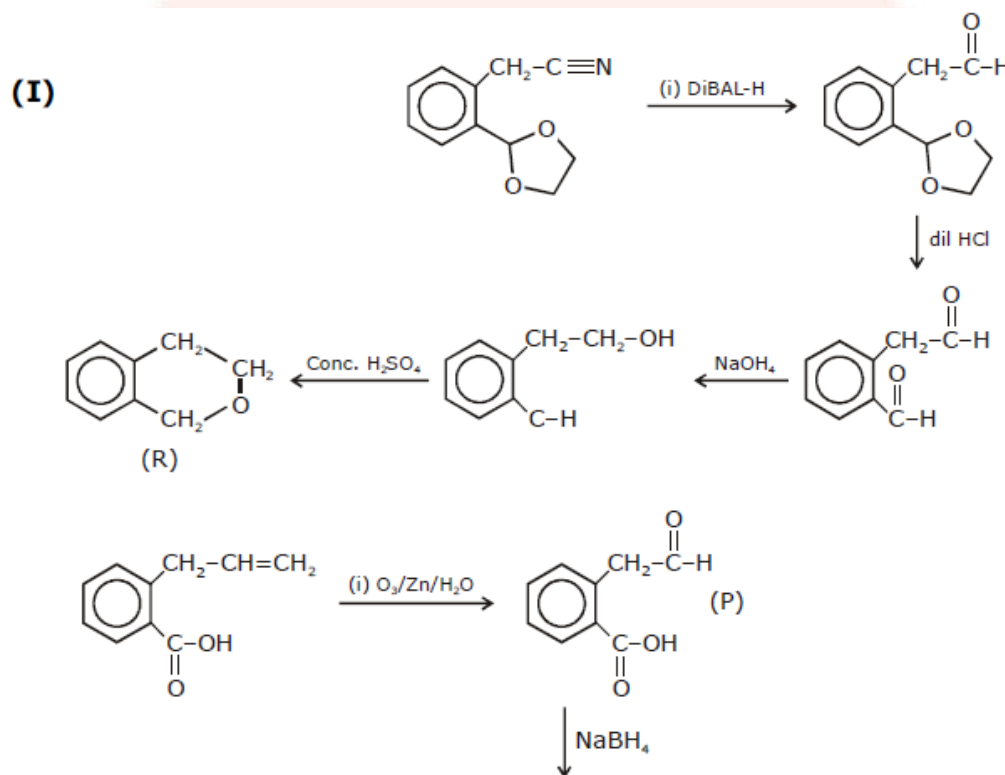
LIST-I		LIST-II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

Which of the following options has correct combination considering List-I and List-II?

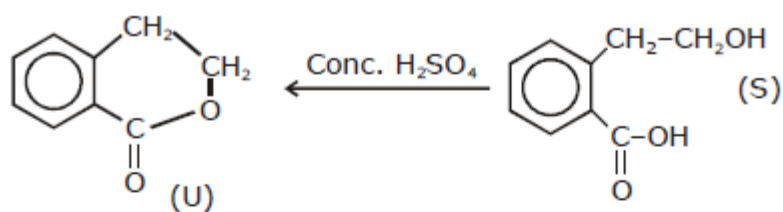
- (1) (I), (Q), (T), (U) (2) (II), (P), (S), (U)
(3) (II), (P), (S), (T) (4) (I), (S), (Q), (R)

Soln.: [2]

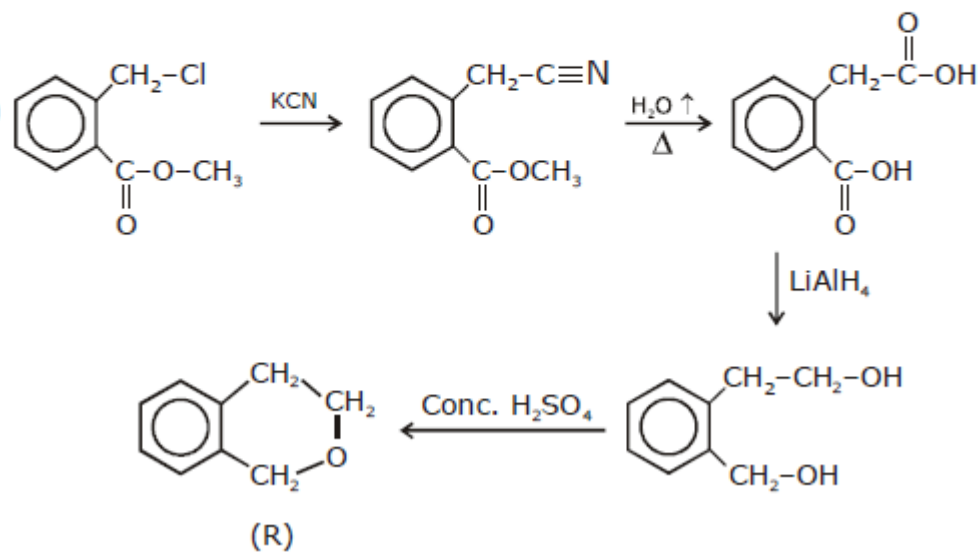
The above reactions as follows:



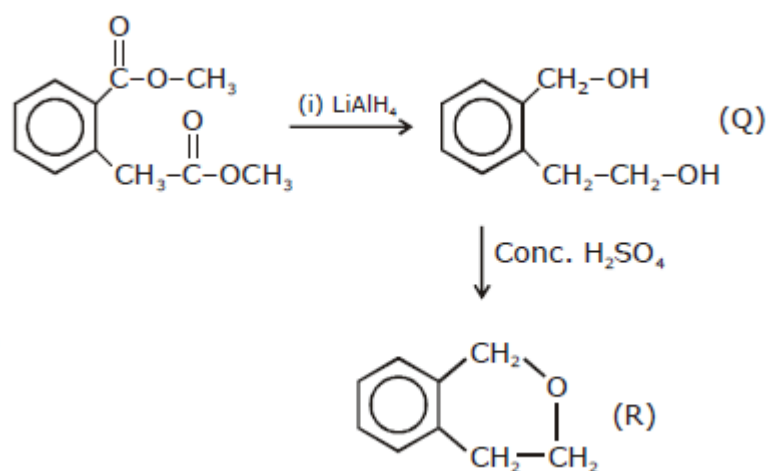
(II)



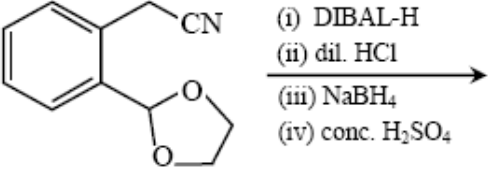
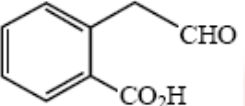
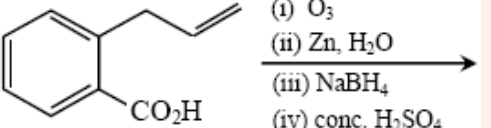
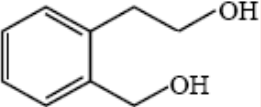
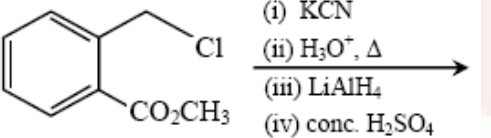
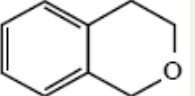
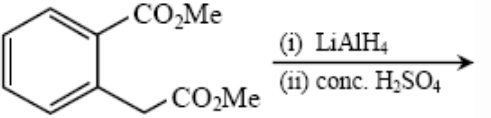
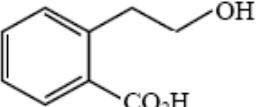
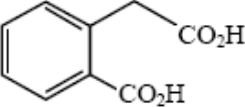
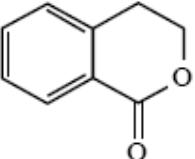
(III)



(IV)



4. List-I includes starting materials and reagents of selected chemical reactions. List-II gives structures of compounds that may be formed as intermediate products and/or final products from the reactions of List-I.

LIST-I		LIST-II	
(I)		(P)	
(II)		(Q)	
(III)		(R)	
(IV)		(S)	
		(T)	
		(U)	

(1) (IV), (Q), (R)

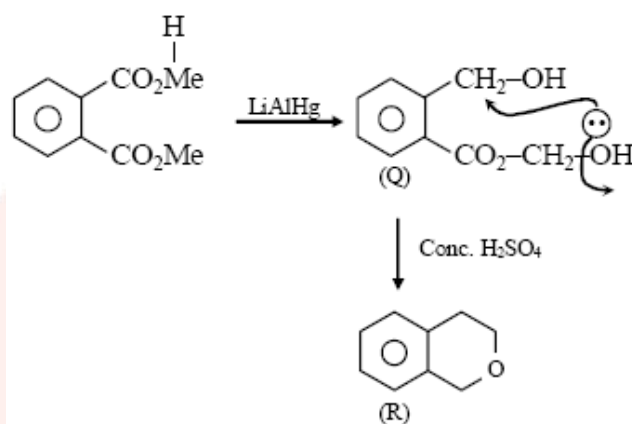
(2) (III), (S), (R)

(3) (IV), (Q), (U)

(4) (III), (T), (U)

Soln.: [1]

The above reactions as follows:



JEE Advanced Exam 2019

(Paper & Solution)

PAPER-2

PART-I (MATHEMATICS)

SECTION – 1(Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct Answer(s).
- For each question, choose(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :
 Full Marks: **+4** If only (all) the correct option(s) is (are) chosen.
 Partial Marks: **+3** If all the four options are correct but **ONLY** three options are chosen.
 Partial Marks: **+2** If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
 Partial Marks: **+1** If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
 Zero Marks: **0** If none of the option is chosen (i.e. the question is unanswered).
 Negative Marks: **–1** In all other cases.

- For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 marks; choosing ONLY (B) will get +1 marks; choosing ONLY (D) will get +1 marks; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -1 mark

1. For $a \in \mathbb{R}, |a| > 1$, let $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$

Then the possible value(s) of a is/are

- (1) 8 (2) -9 (3) -6 (4) 7

Soln.: [1, 2]

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{\sqrt[3]{1} + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) &= \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^{1/3}}{n^{7/3} \left(\sum_{r=1}^n \frac{1}{(an+r)^2} \right)} \\ &= \lim_{n \rightarrow \infty} \left(\frac{\sum_{r=1}^n \left(\frac{r}{n} \right)^{1/3}}{\sum_{r=1}^n \frac{1}{(a + r/n)^2}} \right) \end{aligned}$$

$$\text{Now, } \frac{\int_0^1 x^{1/3} dx}{\int_0^1 \frac{1}{(a+x)^2} dx} = 54 \Rightarrow \frac{\frac{3}{4} \left| x^{4/3} \right|_0^1}{-\left| \frac{1}{a+x} \right|_0^1} = 54 \Rightarrow \frac{\frac{3}{4}}{\frac{1}{a(a+1)}} = 54$$

$$\Rightarrow a(a+1) = 72 \Rightarrow a^2 + a - 72 = 0 \Rightarrow a = -9, 8$$

2. Let $f(x) = \frac{\sin \pi x}{x^2}, x > 0$

Let $x_1 < x_2 < x_3 < \dots x_n < \dots$ be all the points of local maximum of f and

$y_1 < y_2 < y_3 < \dots y_n < \dots$ be all the points of local minimum of f .

Then which of the following options is/are correct?

(1) $x_n \in \left(2n, 2n + \frac{1}{2}\right)$ for every n

(2) $x_1 < y_1$

(3) $|x_n - y_n| > 1$ for every n

(4) $x_{n+1} - x_n > 2$ for every n

Soln.: [1, 3, 4]

$$f(x) = \frac{\sin \pi x}{x^2}$$

$$\Rightarrow f'(x) = \frac{\pi x^2 \cos \pi x - 2x \sin \pi x}{x^4}$$

$$\Rightarrow f'(x) = \frac{2x \cos \pi x \left(\frac{\pi x}{2} - \tan \pi x \right)}{x^4}$$

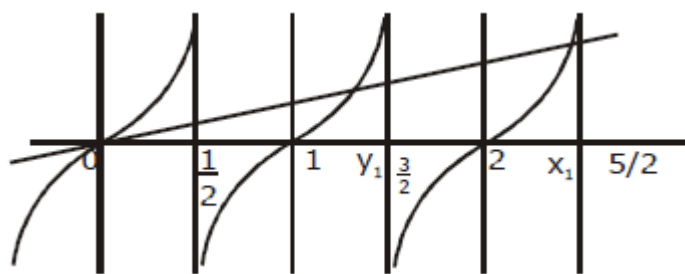
$$\Rightarrow f'(x) = \frac{2 \cos \pi x \left(\frac{\pi x}{2} - \tan \pi x \right)}{x^3}$$

For maxima & minima, $f'(x) = 0$

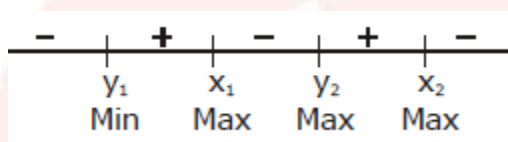
$$\Rightarrow \cos \pi x = 0 \text{ or } \pi x/2 = \tan \pi x$$

$$\Rightarrow \pi x = \frac{(2n+1)\pi}{2} \text{ or } \frac{\pi x}{2} = \tan \pi x$$

$$\Rightarrow x = \frac{(2n+1)}{2}, n \in I$$



$f'(x)$



3. Let $P_1 = I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, $P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$,
 $P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ and $X = \sum_{k=1}^6 P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$

Where P_k^T denotes the transpose of the matrix P_k . Then which of the following options is/are correct?

(1) X is a symmetric matrix

(2) If $X \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \alpha \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, then $\alpha = 30$

(3) The sum of diagonal entries of X is 18

(4) $X - 30I$ is an invertible matrix

Soln.: [1, 2, 3]

We have,

$$P_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, P_3 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, P_4 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, P_5 = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix},$$

$$P_6 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$X = \sum P_k \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix} P_k^T$$

$$\text{Let } A = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\text{then } A^T = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow A = A^T$$

$$\text{Now, } X^T = (P_1 A P_1^T + P_2 A P_2^T + \dots + P_6 A P_6^T)^T = X$$

So X is a symmetric matrix.

$$\text{Let } Q = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$XQ = P_1 A P_1^T Q + P_2 A P_2^T Q + \dots + P_6 A P_6^T Q$$

$$= P_1 A Q + P_2 A Q + \dots + P_6 A Q \quad \left[\because P_1^T Q = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = Q \right]$$

$$= (P_1 + P_2 + \dots + P_6) A Q,$$

$$= \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} 6 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 30 \\ 30 \\ 30 \end{bmatrix} = 30Q \therefore AQ = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 0 & 2 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \\ 6 \end{bmatrix}$$

$$\Rightarrow XQ = 30Q \Rightarrow (X - 30I) Q = 0$$

So, $|X - 30I| = 0$ has non-trivial solution.

So $|X - 30I|$ is not invertible.

Where trace $(P_k A P_k^T) = 3$

$$\Rightarrow \text{Trace } X = 3 \times 6 = 18.$$

4. Let $x \in \mathbb{R}$ and let $P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$, $Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix}$ and $R = PQP^{-1}$. Then

which of the following options is/are correct?

(1) For $x = 1$, there exists a unit vector $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$ for which a

$$R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

(2) There exists a real number x such that $PQ = QP$

(3) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in \mathbb{R}$

(4) For $x = 0$, if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then $a + b = 5$

Soln.: [1, 4]

We have, $R = PQP^{-1}$

$$\Rightarrow \det(R) = \det(PQP^{-1}) = (\det P) (\det Q) \left(\frac{1}{\det P} \right)$$

$$= \det Q$$

$$\det(Q) = \begin{vmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{vmatrix}$$

$$= 2(24) - x(0) + x(-4x)$$

$$= 48 - 4x^2$$

$$\Rightarrow \det R = \det Q = 48 - 4x^2$$

$$(i) \text{ for } x = 1, \det(R) = 48 - 4x^2 = 48 - 4 = 44 \neq 0$$

$$\Rightarrow (\det R) \neq 0$$

$$\therefore \text{ for equation } R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

We will have trivial solution

$$\Rightarrow \alpha = \beta = \gamma = 0$$

$$(2) PQ = QP$$

$$PQP^{-1} = Q$$

$$R = Q$$

No value of x.

$$(3) \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8 = (40 - 4x^2) + 8 = 48 - 4x^2 = \det R \quad \forall x$$

$$\in R$$

$$\text{For } x = 0$$

$$PQ = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 8 & 12 \\ 0 & 0 & 18 \end{bmatrix} \text{ and } P^{-1} = \begin{bmatrix} 6 & -3 & 0 \\ 0 & 3 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

$$R = \begin{bmatrix} 2 & 1 & 2/3 \\ 0 & 4 & 4/3 \\ 0 & 0 & 6 \end{bmatrix}$$

Now $R = PQP^{-1}$

$$= \begin{bmatrix} 2 & 4 & 6 \\ 0 & 8 & 12 \\ 0 & 0 & 18 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 6 & -3 & 0 \\ 0 & 3 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

We have, $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$

$$(R - 6I) \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 0$$

$$\Rightarrow -4 + a + \frac{2b}{3} = 0 \text{ and } -2a + \frac{4b}{3} = 0$$

$$\Rightarrow a = 2 \text{ and } b = 3$$

$$a + b = 5$$

5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function. We say that f has
PROPERTY 1 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite, and
PROPERTY 2 if $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h^2}$ exists and is finite.

Then which of the following options is/are correct?

- (1) $f(x) = x^{2/3}$ has PROPERTY 1
- (2) $f(x) = x|x|$ has PROPERTY 2
- (3) $f(x) = \sin x$ has PROPERTY 2
- (4) $f(x) = |x|$ has PROPERTY 1

Soln.: [1, 4]

(1) We have, $f(x) = x^{2/3}$

$$= \lim_{h \rightarrow 0} \frac{h^{2/3} - 0}{\sqrt[3]{|h|}} = 0 \text{ (finite)}$$

(2) We have $f(x) = x|x|$

$$\lim_{h \rightarrow 0} \frac{h|h| - 0}{h^2} = \lim_{h \rightarrow 0} \frac{|h|}{h}$$

$$\left. \begin{array}{l} \text{Case - I: } \lim_{h \rightarrow 0^+} \frac{|h|}{h} = 1 \\ \text{Case - II: } \lim_{h \rightarrow 0^-} \frac{|h|}{h} = -1 \end{array} \right\} \rightarrow \text{limit does not exist}$$

(3) We have, $f(x) = \sin x$

$$= \lim_{h \rightarrow 0} \frac{\sin h - 0}{h^2} \rightarrow \text{limit does not exist}$$

(4) We have $f(x) = |x|$

$$= \lim_{h \rightarrow 0} \frac{|h| - 0}{\sqrt{|h|}} = 0 \rightarrow \text{finite}$$

So, correct option is (1, 4)

6. Three lines

$L_1: \vec{r} = \lambda \hat{i}, \lambda \in \mathbb{R}$, $L_2: \vec{r} = \hat{k} + \mu \hat{j}, \mu \in \mathbb{R}$ and $L_3: \vec{r} = \hat{i} + \hat{j} + v \hat{k}, v \in \mathbb{R}$ are given. For which point(s) Q on L_2 can we find a point P on L_1 and a point R on L_3 so that P, Q and R are collinear?

(1) $\hat{k} - \frac{1}{2} \hat{j}$

(2) $\hat{k} + \frac{1}{2} \hat{j}$

(3) $\hat{k}$

(4) $\hat{k} + \hat{j}$

Soln. [1, 2]

Let $P \equiv (\lambda, 0, 0)$, $Q(0, \mu, 1)$, $R(1, 1, v)$

P, Q, R are collinear $\overrightarrow{PQ} = k\overrightarrow{PR}$

$$\frac{\lambda - 0}{\lambda - 1} = \frac{0 - \mu}{0 - 1} = \frac{0 - 1}{0 - v}$$

$$\Rightarrow \mu = \frac{1}{v} \text{ and } \lambda = \lambda\mu - \mu$$

$$\Rightarrow v = \frac{1}{\mu} \text{ and } \lambda = \frac{\mu}{\mu - 1}$$

$$\Rightarrow \mu \neq 0, \mu \neq 1$$

$$\Rightarrow Q \neq \hat{k}, Q \neq \hat{k} + \hat{j}$$

7. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x - 1)(x - 2)(x - 5)$. Define

$$F(x) = \int_0^x f(t) dt, \quad x > 0$$

Then which of the following options is/are correct?

- (1) F has a local minimum at $x = 1$
- (2) F has two local maxima and one local minimum in $(0, \infty)$
- (3) F has a local maximum at $x = 2$
- (4) $F(x) \neq 0$ for all $x \in (0, 5)$

Soln.: [1, 3, 4]

We have,

$$f(x) = (x - 1)(x - 2)(x - 5)$$

$$F(x) = \int_0^x f(t) dt, \quad x > 0$$

$$F'(x) = f(x) = (x - 1)(x - 2)(x - 5), \quad x > 0$$

$x = 1, 5$ is point of local minima for $x > 0$

$x = 2$ is point of local maxima for $x > 0$

$$f(x) = x^3 - 8x^2 + 17x - 10$$

$$\Rightarrow F(x) = \int_0^x (t^3 - 8t^2 + 17t - 10) dt$$

$$F(x) = \frac{x^4}{4} - \frac{8x^3}{3} + \frac{17x^2}{2} - 10x$$

$$\text{Also } F(2) = \int_0^2 f(t) dt < 0$$

$$\Rightarrow F(x) < 0 \quad \forall x \in (0, 5)$$

8. For non-negative integers n , let
$$f(x) = \frac{\sum_{k=0}^n \sin\left(\frac{k+1}{n+2}\pi\right) \sin\left(\frac{k+2}{n+2}\pi\right)}{\sum_{k=0}^n \sin^2\left(\frac{k+1}{n+2}\pi\right)}$$

Assuming $\cos^{-1} x$ takes values in $[0, \pi]$ which of the following options is/are correct?

(1) $f(4) = \frac{\sqrt{3}}{2}$

(2) $\sin(7 \cos^{-1} f(5)) = 0$

(3) If $\alpha = \tan(\cos^{-1} f(6))$, then $\alpha^2 + 2\alpha - 1 = 0$

(4) $\lim_{n \rightarrow \infty} f(n) = \frac{1}{2}$

Soln.: [1, 2, 3]

$$f(x) = \frac{\sum_{K=0}^n \left(2 \sin\left(\frac{K+1}{n+2}\pi\right) \sin\left(\frac{K+2}{n+2}\pi\right) \right)}{\sum_{K=0}^n 2 \sin^2\left(\frac{K+1}{n+2}\pi\right)}$$

$$\Rightarrow f(x) = \frac{\sum_{k=0}^n \left(\cos\left(\frac{\pi}{n+2}\right) - \cos\left(\frac{2k+3}{n+2}\pi\right) \right)}{\sum_{k=0}^n \left(1 - \cos\left(\frac{2k+2}{n+2}\pi\right) \right)}$$

$$\begin{aligned}
 &= \frac{(n+1)\cos\left(\frac{\pi}{n+2}\right) - \frac{\cos\left(\frac{n+3}{n+2}\right)\pi \cdot \sin\left(\frac{n+1}{n+2}\right)\pi}{\sin\left(\frac{\pi}{n+2}\right)}}{(n+1) - \frac{\cos\pi \cdot \sin\left(\frac{n+1}{n+2}\right)\pi}{\sin\left(\frac{\pi}{n+2}\right)}} = \frac{(n+2)\cos\left(\frac{\pi}{n+2}\right)}{n+2} \\
 &= \frac{(n+1)\cos\left(\frac{\pi}{n+2}\right) + \cos\left(\frac{\pi}{n+2}\right)}{(n+1)+1} = \cos\left(\frac{\pi}{n+2}\right)
 \end{aligned}$$

Hence, $f(x) = \cos\left(\frac{\pi}{n+2}\right)$

Now, (1) $f(4) = \cos\left(\frac{\pi}{4+2}\right) = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

(2) $\sin\left(7\cos^{-1}\left(\cos\frac{\pi}{7}\right)\right) = \sin\left(7 \times \frac{\pi}{7}\right) \sin\pi = 0$

(3) $\alpha = \tan\left(\cos^{-1}\left(\cos\frac{\pi}{8}\right)\right) = \tan\frac{\pi}{8}$ Also $\tan\frac{\pi}{4} = \frac{2\tan\frac{\pi}{8}}{1 - \tan^2\frac{\pi}{8}} \Rightarrow 1 = \frac{2\alpha}{1 - \alpha^2}$
 $= \alpha^2 + 2\alpha - 1 = 0$

(4) $\lim_{n \rightarrow \infty} f(n) = \lim_{n \rightarrow \infty} \cos\left(\frac{\pi}{n+2}\right) = 1$

SECTION – 2 (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to TWO decimal places.

- Answer to each question will be evaluated according to the following marking scheme.

Full Marks: +3 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

1. Suppose $\det \begin{bmatrix} \sum_{k=0}^n k & \sum_{k=0}^n {}^nC_k k^2 \\ \sum_{k=0}^n {}^nC_k k & \sum_{k=0}^n {}^nC_k 3^k \end{bmatrix} = 0$, holds for some positive integer

n. Then $\sum_{k=0}^n \frac{{}^nC_k}{k+1}$ equals

Soln.: [6.20]

$$\sum_{k=0}^n (k^2 - k + k) {}^nC_k = \sum_{k=0}^n (k-1) \frac{n}{k} \cdot \frac{n-1}{k-1} {}^{n-1}C_{k-2} + \sum_{r=0}^n k \frac{n}{k} {}^{n-1}C_{k-1}$$

$$= n(n-1) \cdot 2^{n-2} + n \cdot 2^{n-1}$$

$$= n \cdot 2^{n-2} [(n-1)+2]$$

$$= n(n+1) 2^{n-2}$$

$$\sum_{k=0}^n k \frac{n}{k} {}^{n-1}C_{k-1} = n 2^{n-1}$$

$$\sum_{k=0}^n 3k {}^{n-1}C_{k-1} = {}^nC_0 + 3({}^nC_1) + (3^2)n_e + \dots 3^n ({}^nC_n)$$

Now $\begin{vmatrix} \frac{n(n+1)}{2} & n(n+1)2^{n-2} \\ n2^{n-1} & 4^n \end{vmatrix} = 0$

$$\begin{vmatrix} 2 & 2^n \\ n2^{n-1} & 4^n \end{vmatrix} = 0 \quad 2^{2n+1} = n \cdot 2^{2n-1}$$

$$= n = 4$$

$$\begin{aligned}\sum_{k=0}^4 \frac{{}^4C_k}{k+1} &= \frac{{}^4C_0}{1} + \frac{{}^4C_1}{2} + \frac{{}^4C_2}{3} + \frac{{}^4C_3}{4} + \frac{{}^4C_4}{5} \\ &= 1 + 2 + 2 + 1 + 1/5 \\ &= \frac{31}{5} \\ &= 6.20\end{aligned}$$

2. Let $\vec{a} = 2\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$ be two vectors. Consider a vector $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, $\beta \in \mathbb{R}$. If projection of $\vec{c}$ on the vector $(\vec{a} + \vec{b})$ is $3\sqrt{2}$, then the minimum value of $(\vec{c} - (\vec{a} \times \vec{b})) \cdot \vec{c}$ equals.....

Soln.: [18]

$$\vec{c} = \alpha(2\hat{i} + \hat{j} - \hat{k}) + \beta(\hat{i} + 2\hat{j} + \hat{k})$$

$$\vec{c} = (2\alpha + \beta)\hat{i} + (\alpha + 2\beta)\hat{j} + (\beta - \alpha)\hat{k}$$

$$\text{Projection of } \vec{c} \text{ on the vector } (\vec{a} + \vec{b}) = \frac{\vec{c} \cdot (\vec{a} + \vec{b})}{|\vec{a} + \vec{b}|}$$

$$\frac{\vec{c} \cdot (\vec{a} + \vec{b})}{|\vec{a} + \vec{b}|} = 3\sqrt{2}$$

$$9(\alpha + \beta) = 18$$

$$\alpha + \beta = 2$$

$$\text{Now, } (\vec{c} - (\vec{a} \times \vec{b})) \cdot \vec{c} = \alpha^2 |\vec{a}|^2 + \beta^2 |\vec{b}|^2 + 2\alpha\beta \vec{a} \cdot \vec{b} - (\vec{a} \times \vec{b}) \cdot \vec{c} = 6(\alpha^2 - 2\alpha + 4)$$

Minimum value = 18

3. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ equals.....

Soln.: [0.5]

$$\text{Let } I = \int_0^{\pi/2} \frac{3\sqrt{\cos \theta} d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5}$$

$$\Rightarrow 2I = 3 \int_0^{\pi/2} \frac{d\theta}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^4}$$

$$\Rightarrow I = \frac{3}{2} \int_0^{\pi/2} \frac{\sec^2 \theta d\theta}{(2 + \sqrt{\tan \theta})^4}$$

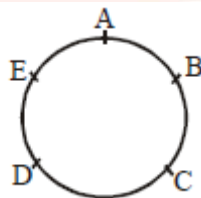
$$\text{Let } 1 + \sqrt{\tan \theta} = t \Rightarrow \frac{1}{2\sqrt{\tan \theta}} \sec^2 \theta d\theta = dt$$

$$I = \frac{3}{2} \int_1^{\infty} \frac{2(t-1)}{t^4} dt = \frac{3}{2} \left[\frac{2}{-2t^2} + \frac{2}{3t^3} \right]_1^{\infty} \left[\begin{array}{ll} \text{when } \theta = 0, & \text{then } t = 1 \\ \text{and } \theta = \frac{\pi}{2}, & \text{then } t = \infty \end{array} \right]$$

$$= \frac{3}{2} \left(\frac{2}{2} - \frac{2}{3} \right) = \frac{1}{2} = 0.5$$

4. Five persons A, B, C, D and E are seated in a circular arrangement. If each of them is given a hat of one of the three colors red, blue and green, then the number of ways of distributing the hats such that the persons seated in adjacent seats get different coloured hats is

Soln.: [30]



Possibilities are

1R, 2B, 2G; 1B, 2R, 2G; or 1G, 2R, 2B

$\Rightarrow$ Total number of ways = ${}^5C_1 \times 2 + {}^5C_1 \times 2 + {}^5C_1 \times 2 = 10 \times 3 = 30$

So total no. of ways = $3 \times 10 = 30$

5. The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals.....

Soln.: [0]

We have, $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$

$$= \sec^{-1}\left(\frac{1}{4}\sum_{K=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{K\pi}{2}\right)\sec\left(\frac{\pi}{2}+\frac{7\pi}{12}+\frac{(K)\pi}{2}\right)\right)$$

$$= \sec^{-1}\left(-\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\operatorname{cosec}\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\right)$$

$$= \sec^{-1}\left[-\frac{1}{4}\sum_{k=0}^{10}\frac{2}{\sin\left(\frac{7\pi}{6}+k\pi\right)}\right]$$

$$= \sec^{-1}\left[-\frac{1}{2}\sum_{k=0}^{10}\frac{1}{(-1)^{k+1}\sin\frac{\pi}{6}}\right]$$

$$= \sec^{-1}(1) = 0$$

6. Let $|X|$ denote the number of elements in a set X . Let $S = \{1, 2, 3, 4, 5, 6\}$ be a sample space, where each elements is equally likely to occur. If A and B are independent events associated with S , then the number of ordered pairs (A, B) such that $1 \leq |B| < |A|$, equals

Soln.: [422]

A and B are independent events

$$\therefore P(A \cap B) = P(A) \cdot P(B)$$

$$\Rightarrow \frac{|A \cap B|}{|S|} = \frac{|A|}{|S|} \cdot \frac{|B|}{|S|}$$

$$\Rightarrow |S| |A \cap B| = |A| \cdot |B|$$

$$\Rightarrow 6 \cdot |A \cap B| = |A| \cdot |B| \quad \dots(i)$$

$$\therefore |B| < |A|$$

So, possible values of $|A|$ can be 2, 3, 4 or 6

$|A| \neq 5$; because $|A| = 5$ cannot satisfy the equation (i)

Case I

If $|A| = 2$ then $|B| = 1$ (it is also not satisfying equation (i))

Case-2 :

If $|A| = 3$ then $|B| = 2$ and $|A \cap B| = 1$

$$\text{Number of ways} = {}^6C_3 \cdot {}^3C_1 \cdot {}^3C_1 = 180$$

Case-3 :

If $|A| = 4$ then $|B| = 3$ and $|A \cap B| = 2$

$$\text{Number of ways} = {}^6C_4 \cdot {}^4C_2 \cdot {}^2C_1 = 180$$

Case-4 :

If $|A| = 6$ then $|B|$ may be any number between 1 to 5

$$\text{Number of ways} = {}^6C_6 \cdot [{}^6C_1 + {}^6C_2 + \dots + {}^6C_5] = 62$$

$$\text{Total number of ordered pairs } (A, B) = 180 + 180 + 62 = 422$$

SECTION – 3 (Maximum Marks: 12)

- This section contains **Two (02)** List-Match sets.
- Each List-Match set has **TWO (02)** Multiple Choice Questions.
- Each List-Match set has two lists : **List-I** and **List-II**
- **List-I** has **Four** entries (I), (II), (III) and (IV) and List-II has Six entries (P), (Q), (R), (S), (T) and (U).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- For each question, marks will be awarded according to the following marking scheme :
Full Marks: +3 If **ONLY** the option corresponding to the correct matching is chosen.
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered.)
Negative Marks: –1 In all other cases.

Answer the following by appropriately matching the lists based on the information given in the paragraph.

1. Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order:

$$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$$

$$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}$$

List-I contains the sets X, Y, Z and W. List-II contains some information regarding these sets.

List-I		List-II	
(I)	X	(P)	$\supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$
(II)	Y	(Q)	an arithmetic progression
(III)	Z	(R)	NOT an arithmetic progression
(IV)	W	(S)	$\supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$
		(T)	$\supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$
		(U)	$\supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$

Which of the following is the only CORRECT combination?

(1) (II), (Q), (T)

(2) (I), (P), (R)

(3) (II), (R), (S)

(4) (I), (Q), (U)

Soln.: [1]

$$X = \{x : f(x) = 0\}$$

$$f(x) = 0 \Rightarrow \sin(\pi \cos x) = 0$$

$$\Rightarrow \pi \cos x = n\pi, n \in I$$

$$\Rightarrow \cos x = n$$

So $\cos x = -1, 0, 1$

$$x = \left\{ \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \dots \right\} \quad X = \left\{ n, \frac{\pi}{2}, n \in I \right\}$$

Also, $Y = \{x : f'(x) = 0\}$

Now, $f'(x) = 0$

$$\Rightarrow f'(x) = -\cos(\pi \cos x) \pi \sin x = 0$$

$$\Rightarrow \cos(\pi \cos x)(-\pi \sin x) = 0$$

$$\Rightarrow \cos(\pi \cos x) = 0 \text{ or } (-\pi \sin x) = 0$$

$$\Rightarrow \pi \cos x = (2n+1)\frac{\pi}{2} \text{ or } x = n\pi$$

$$\Rightarrow \cos x = (2n+1)\frac{1}{2} \text{ or } x = n\pi$$

$$\Rightarrow x = n\pi \pm \frac{\pi}{3} \text{ or } x = n\pi$$

$$Y = \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \frac{3\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{6\pi}{3}, \frac{\pi}{3}, \dots \right\}$$

$$Z = \{x : g(x) = 0\}$$

$$g(x) = 0 \Rightarrow \cos(2\pi \sin x) = 0$$

$$\Rightarrow 2\pi \sin x = (2n+1)\frac{\pi}{2}, n \in I$$

$$\Rightarrow \sin x = \frac{2n+1}{4}$$

$$\Rightarrow \sin x = \pm \frac{1}{4}, \pm \frac{3}{4}$$

$$\Rightarrow Z = \left\{ n\pi \pm \sin^{-1} \frac{1}{4}, n\pi \pm \sin^{-1} \frac{3}{4}, n \in I \right\}$$

$$W = \{x : g'(x) = 0\}$$

$$g'(x) = 0 \Rightarrow -\sin(2\pi \sin x)(2\pi \cos x) = 0$$

$$\Rightarrow \sin(2\pi \sin x) = 0 \text{ or } 2\pi \cos x = 0$$

$$\Rightarrow 2\pi \sin x = n\pi \text{ or } x = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow \sin x = \frac{n}{2} \text{ or } x = (2n+1)\frac{\pi}{2}, n \in I$$

$$\Rightarrow \sin x = 0, \pm \frac{1}{2}, \pm 1 \text{ or } x = (2n+1)\frac{\pi}{2}$$

$$W = \left\{ n\pi, (2n+1)\frac{\pi}{2}, n\pi \pm \frac{\pi}{6}, n \in I \right\}$$

Option (1) is correct.

2. Let $f(x) = \sin(\pi \cos x)$ and $g(x) = \cos(2\pi \sin x)$ be two functions defined for $x > 0$. Define the following sets whose elements are written in the increasing order:

$$X = \{x : f(x) = 0\}, \quad Y = \{x : f'(x) = 0\}$$

$$Z = \{x : g(x) = 0\}, \quad W = \{x : g'(x) = 0\}$$

List-I contains the sets X, Y, Z and W. List-II contains some information regarding these sets.

List-I		List-II	
(I)	X	(P)	$\supseteq \left\{ \frac{\pi}{2}, \frac{3\pi}{2}, 4\pi, 7\pi \right\}$
(II)	Y	(Q)	an arithmetic progression
(III)	Z	(R)	NOT an arithmetic progression
(IV)	W	(S)	$\supseteq \left\{ \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6} \right\}$
		(T)	$\supseteq \left\{ \frac{\pi}{3}, \frac{2\pi}{3}, \pi \right\}$
		(U)	$\supseteq \left\{ \frac{\pi}{6}, \frac{3\pi}{4} \right\}$

Which of the following is the only CORRECT combination?

- (1) (III), (P), (Q), (U) (2) (IV), (P), (R), (S)
 (3) (III), (R), (U) (4) (IV), (Q), (T)

Soln.: [2]

$$f(x) = 0 \Rightarrow \sin(\pi \cos x) = 0$$

$$\Rightarrow \pi \cos x = n\pi \Rightarrow \cos x = n$$

$$\Rightarrow \cos x = -1, 0, 1 \Rightarrow x = (n\pi, (2n+1)\pi/2)$$

$$= (n\pi/2, n)$$

$$f'(x) = 0 \Rightarrow \cos(\pi \cos x) (-\pi \sin x) = 0$$

$$\Rightarrow \pi \cos x = (2n+1)\pi/2 \text{ or } x = n\pi$$

$$\Rightarrow \cos x = -1, 0, 1 \Rightarrow x = (n\pi(2n+1)\frac{\pi}{2}) = \left(n\frac{\pi}{2}, n \in \mathbb{Z}\right)$$

$$f'(x) = 0 \Rightarrow \cos(\pi \cos x) (-\pi \sin x) = 0$$

$$\Rightarrow \cos x = n + \frac{1}{2} \text{ or } x = n\pi$$

$$\Rightarrow \cos x = \pm \frac{1}{2} \text{ of } x = n\pi$$

$$\Rightarrow y = \left\{ 2n\pi \pm \frac{\pi}{3}, 2n\pi + \frac{2\pi}{3}, n\pi, n \in \mathbb{Z} \right\}$$

$$g(x) = 0 \Rightarrow \cos(2x \sin x) = 0 \Rightarrow 2\pi \sin x = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow \sin x = \frac{2n+1}{2} - \frac{1}{2} + \frac{3}{2}$$

$$\Rightarrow \cos x = n + \frac{1}{2} \text{ or } x = n\pi \Rightarrow \cos x = \pm \frac{1}{2} \text{ or } x = n\pi$$

$$\Rightarrow y = \left\{ 2n\pi \pm \frac{\pi}{3}, 2n\pi + \frac{2\pi}{3}, n\pi, n \in \mathbb{Z} \right\}$$

$$g(x) = 0 \Rightarrow \cos(2\pi \sin x) = 0$$

$$\Rightarrow 2\pi \sin x = (2n+1)\frac{\pi}{2} \Rightarrow \sin x = \frac{2n+1}{4} = \pm \frac{1}{4} \pm \frac{3}{4}$$

$$\Rightarrow Z = \left\{ n\pi \pm \sin^{-1}\frac{1}{4}, n\pi \pm \sin^{-1}\frac{3}{4}, n \in \mathbb{Z} \right\}$$

$$g'(x) = 0 \Rightarrow -\sin(2\pi \sin x)(2\pi \cos x) = 0$$

$$\Rightarrow 2\pi \sin x = n\pi \text{ or } x = (2n+1)\frac{\pi}{2} \Rightarrow \sin x = \frac{n}{2} = 0, \pm \frac{1}{2}, \pm 1 \text{ or } x = (2n+1)\frac{\pi}{2}$$

$$\Rightarrow W = \left\{ n\pi, (2n+1)\frac{\pi}{2}, n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z} \right\}$$

3. Let the circles $C_1 : x^2 + y^2 = 9$ and $C_2 : (x-3)^2 + (y-4)^2 = 16$, intersect at the points X and Y. Suppose that another circle

$C_3 : (x-h)^2 + (y-k)^2 = r^2$ satisfies the following conditions.

- (i) center of C_3 is collinear with the centers of C_1 and C_2 ,
- (ii) C_1 and C_2 both lie inside C_3 , and
- (iii) C_3 touches C_1 at M and C_2 at N.

Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2 = 8\alpha y$.

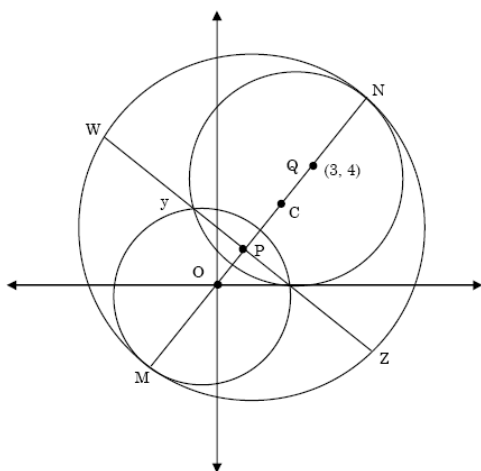
There are some expressions given in the List-I whose values are given in List-II below:

List-I		List-II	
(I)	$2h + k$	(P)	6
(II)	$\frac{\text{Length of ZW}}{\text{Length of XY}}$	(Q)	$\sqrt{6}$
(III)	$\frac{\text{Area of triangle MZN}}{\text{Area of triangle ZMW}}$	(R)	$\frac{5}{4}$
(IV)	α	(S)	$\frac{21}{5}$
		(T)	$2\sqrt{6}$
		(U)	$\frac{10}{3}$

Which of the following is the only INCORRECT combination?

- (1) (IV), (U)
- (2) (I), (P)
- (3) (IV), (S)
- (4) (III), (R)

Soln.: [3]



(I) Diameter of circle C_3 will lie on line $MN \Rightarrow 2r = MN$
 $= MO + OQ + QN$
 $2r = 3 + \sqrt{3^2 + 4^2} + 4 = 3 + 5 + 4 = 12$
 $\therefore r = 6$

Centre C of circle C_3 lies on line $y = \frac{4}{3}x$

Let co-ordinates of $C(h, k)$ where $k = \frac{4h}{3}$

$$OC = MC - OM$$

$$OC = 6 - 3 = 3$$

$$\therefore \sqrt{h^2 + \frac{16}{9}h^2} = 3 \Rightarrow \sqrt{\frac{25h^2}{9}} = 3 \Rightarrow \frac{5h}{3} = 3$$

$$\Rightarrow h = \frac{9}{5}$$

$$\Rightarrow k = \frac{4}{3}h = \frac{4}{3} \times \frac{9}{5} = \frac{12}{5}$$

$$\therefore 2h + k = \frac{18}{5} + \frac{12}{5} = 6$$

(II) Equation of line ZW

$$C_1 - C_2 = 0$$

$$\Rightarrow 6x + 8y - 19 = 0$$

$$\Rightarrow 3x + 4y = 9$$

Perpendicular distance of ZW from (0, 0) is

$$= \left| \frac{9}{\sqrt{9+16}} \right| = \frac{9}{5}$$

$$\text{Length of XY} = 2\sqrt{9 - \left(\frac{9}{5}\right)^2} = \frac{24}{5}$$

Perpendicular distance of ZW from $C_3\left(\frac{9}{5}, \frac{12}{5}\right)$ is

$$\left| \frac{3\left(\frac{9}{5}\right) + 4\left(\frac{12}{5}\right) - 9}{\sqrt{9+16}} \right| = \frac{6}{5}$$

$$\text{Length of ZW} = 2\sqrt{36 - \left(\frac{6}{5}\right)^2} = 2\sqrt{36 - \frac{36}{25}} = 2\sqrt{36\left(1 - \frac{1}{25}\right)} = \frac{24\sqrt{6}}{5}$$

$$\therefore \frac{\text{length of ZW from } \left(\frac{9}{5}, \frac{12}{5}\right)}{\text{length of XY, from } (0, 0)} = \sqrt{6}$$

$$\text{(III) Area of } \Delta MZN = \frac{1}{2} ZP \times MN = \frac{1}{2} NM \left(\frac{1}{2} ZW \right)$$

$$\text{Area of } \Delta ZMW = \frac{1}{2} ZW \times MP = \frac{1}{2} NM \left(\frac{1}{2} ZW \right)$$

$$\therefore \frac{\text{Area of } \Delta MZN}{\text{Area of } \Delta ZMW} = \frac{5}{4}$$

(IV) Slope of tangent to C_1 at $M = -\frac{3}{4}$

$$\therefore \text{Equation of tangent } y = mx - 3\sqrt{1+m^2}$$

$$y = -\frac{3}{4}x - 3\sqrt{1+\frac{9}{16}}$$

$$\Rightarrow y = -\frac{3x}{4} - \frac{15}{4} \quad \dots\dots (i)$$

Tangent to parabola $x^2 = 8\alpha y$ is

$$\Rightarrow y = mx - 2\alpha m^2 \quad \dots\dots(ii)$$

Compare (i) & (ii)

$$m = -\frac{3}{4} \text{ and } 2\alpha m^2 = +\frac{15}{4}$$

$$\Rightarrow 2\alpha \left(\frac{9}{16}\right) = \frac{15}{4}$$

$$\Rightarrow \alpha = \frac{10}{3}$$

So, correct option (3) (IV) - (S)

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Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and C_3 be a tangent to the parabola

$$x^2 = 8\alpha y.$$

There are some expressions given in the List-I whose values are given in List-II below:

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(IV)	α	(S)	$\frac{21}{5}$

(1) (II), (Q)	(2) (II), (T)
(3) (I), (S)	(4) (I), (U)

Refer solution of above Question

So, $\text{II} \rightarrow \text{Q}$